DLS Buddy — Theory and Equations Guide

Applies to v1.17.0 and later

Esam Alfalah

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Equations are numbered (1), (2), … and rendered in LaTeX; an in-text reference like "Eq. 2" resolves to the numbered display equation, and citations resolve to the numbered reference list at the end.

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1. Physics and Calibration

1.1 Scattering Vector

$$q = \frac{4\pi n}{\lambda} \sin\left(\frac{\theta}{2}\right) \tag{1}$$

where λ is the vacuum wavelength of the laser, n is the solvent refractive index at that wavelength and temperature, and θ is the scattering angle. This is the standard definition for solution scattering [1].

1.2 Stokes–Einstein Relation

The hydrodynamic radius follows from the translational diffusion coefficient $D = \Gamma/q^2$ (Γ is the field-autocorrelation decay rate) via the **Stokes–Einstein relation** [2]:

$$R_h = \frac{k_B T}{6\pi\eta D} \tag{2}$$

where k_B is the Boltzmann constant, T the absolute temperature, and η the solvent dynamic viscosity. A single-angle measurement gives the **apparent** R_h; extrapolation to $q\to 0$ and $c\to 0$ is handled in the analysis module.

1.3 Refractive-Index Correction

When the sample solvent differs from the calibration standard, the excess Rayleigh ratio is corrected by

$$f = \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}}\right)^2 \tag{3}$$

applied as $\Delta R_\theta = R_\theta{}^{\text{measured}} \cdot f$. When solvent = standard, $f = 1$ exactly. It is applied whenever the solvent differs from the calibration liquid. The fixed squared form follows from refraction of the

observed scattering volume and solid angle at the cell wall, and holds for rectangular and cylindrical cells alike [3]. If n_{standard} is not supplied the correction cannot be formed; the program then treats the run as **uncalibrated** — M_{w} and the absolute A_2 flagged, R_{g} and $2A_2M_{\text{w}}$ unaffected — rather than silently using $f = 1$.

1.4 Optical Constant

The prefactor in the Zimm/Debye/Berry equations (vertically polarized geometry):

$$K = \frac{4\pi^2 n^2 \left(\mathrm{d}n/\mathrm{d}c\right)^2}{N_A \lambda^4} \tag{4}$$

with N_{A} the Avogadro number. Units are mol·mL·g^{-2}·cm^{-1}, consistent with c in g/mL and R_θ in cm^{-1}.

> **Important.** $K \propto (\mathrm{d}n/\mathrm{d}c)^2$, so the refractive-index increment enters the molecular weight **squared**. For a low-contrast system a small $\mathrm{d}n/\mathrm{d}c$ makes the scattering weak and amplifies its error, which is why the program takes it as a hand-entered value rather than proposing one.

When no $\mathrm{d}n/\mathrm{d}c$ exists. Some samples have none to give: no increment has been published for the polymer, or — for a particle dispersion — the quantity is not defined at all. The program accepts a **blank** $\mathrm{d}n/\mathrm{d}c$ rather than requiring a number, because K multiplies the *whole* ordinate $Kc/\Delta R$ and therefore cancels in every quantity extracted as a **ratio of fitted coefficients**. Concretely, with $\mathrm{d}n/\mathrm{d}c$ absent the program sets $K = 1$ — an arbitrary optical scale, recorded on the result and disclosed wherever its dependent values appear — and then:

Unaffected by an absent $\mathrm{d}n/\mathrm{d}c$	On an arbitrary scale, and flagged
every R_{g} (Guinier, Debye, Zimm, Berry, form-factor)	absolute M_{w}
form-factor radii and the ellipsoid axial ratio	absolute A_2
the calibration-free $2{\cdot}A_2{\cdot}M_{\text{w}}$ product (Eq. 69)	the single-angle apparent M_{w}

Guinier's R_{g} is the strongest case: it fits $\ln \Delta R$ against q^2, and $\mathrm{d}n/\mathrm{d}c$ does not enter ΔR at all, so its R_{g} is **bit-identical** with any increment and with none. The ordinate fits cancel K rather than never seeing it, so they agree to within floating-point rounding.

This is the same treatment an **uncalibrated** run receives (§1.5), and for the same reason — an unknown positive factor on the ordinate. The two absences are independent and are reported independently: a fully calibrated run with no $\mathrm{d}n/\mathrm{d}c$ still yields an M_{w} wrong by orders of magnitude, and the program says so rather than implying that calibrating would fix it. Nothing is guessed: a blank $\mathrm{d}n/\mathrm{d}c$ stays blank (invariant: the increment is never proposed), and $K = 1$ is a disclosed placeholder, not a proposed physical value.

A $\mathrm{d}n/\mathrm{d}c$ of **zero** is refused outright. It is not an "unknown" — it asserts that $K = 0$, which collapses the entire ordinate to zero and makes every fit meaningless. **Negative** increments are accepted: K depends on the square, so the sign cancels, and a solute less refractive than its solvent genuinely has one.

One further case joins the blank one. Because K goes as the **square** of $\mathrm{d}n/\mathrm{d}c$ (and as n^2 and λ^{-4}), a value that is legal, finite and non-zero can still leave K unusable — too small to represent, or too large. The program checks K itself rather than the increment, and when it is not a usable positive number it takes the same disclosed $K = 1$ path, saying so in the note. Checking the result rather than the input is deliberate: the boundary depends on three parameters at once, so a test written on $\mathrm{d}n/\mathrm{d}c$ alone could be wrong in three different ways.

1.5 Toluene Rayleigh Ratio (geometry-aware)

All SLS calibration uses the temperature- and geometry-corrected toluene Rayleigh ratio rather than the value stored in instrument software (which goes stale after a re-alignment). The base quantity is the polarized ratio R_{VV}, temperature-corrected and then converted to the measurement geometry:

$$R_{VV}(T) = R_{VV}(25^{\circ}\mathrm{C})\,[\,1 + \alpha\,(T - 25)\,], \qquad \alpha = +0.00421\ ^{\circ}\mathrm{C}^{-1} \tag{5}$$

$$R_{VU} = R_{VV}\,(1 + \rho_v), \qquad R_{VH} = R_{VV}\,\rho_v \tag{6}$$

where $\rho_{\mathrm{v}} = I_{\mathrm{VH}}/I_{\mathrm{VV}}$ is the depolarization ratio, $\rho_{\mathrm{v}}(T) = 0.346 - 0.00135\ (T - 25)$ unless overridden.

Wavelength	R_{VV} at 25 °C (cm^{-1})	Source
532 nm	2.34×10^{-5}	[4]
660 nm	8.456×10^{-6}	[5], Table 2

> **Important.** The temperature coefficient α is **positive** — the Rayleigh ratio *rises* with temperature. The geometry (VV / VU / VH) must match how the calibrant was measured, or the absolute scale is wrong by $\approx (1 + \rho_{\mathrm{v}}) \approx 1.35$. An instrument with a vertical laser and **no analyzer** is **VU** geometry. Example: $R_{\mathrm{VU}}(532\ \mathrm{nm}, 25\ ^{\circ}\mathrm{C}) = 2.34 \times 10^{-5} \times (1 + 0.346) \approx 3.15 \times 10^{-5}\ \mathrm{cm}^{-1}$.
>
> **Other wavelengths** need a separately sourced R_{VV}, not an interpolation: the wavelength dependence is steep — toluene's Rayleigh ratio scales as $\approx \lambda^{-4.17}$, steeper than the λ^{-4} Rayleigh law, reducing to $\approx \lambda^{-4}$ only once the $n(\lambda)$ dispersion is separated out [6]. A laser wavelength within 2 nm of a tabulated one is served that entry **unscaled**, and the program says so beside k_{c} with the approximate size of the substitution on the $\lambda^{-4.17}$ law (≈ 1.6 % per 2 nm at 532 nm); the 2 nm tolerance is a convenience, not a physical criterion.

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2. Data-Quality and Cross-Analysis Theory

2.1 Intensity-trace diagnostics

The trace diagnostics answer two different questions, and keeping them apart matters when you read the plot. The outlier band asks whether any single *sample* stands out from the rest of the

trace; the Fano factor asks whether the *whole trace* fluctuates more than photon statistics permit. A trace can be far from shot-noise-limited and still flag no outliers at all.

Outlier band. A sample is flagged when it lies outside a band centered on the trace's median and scaled by the trace's own scatter [7]:

$$\text{flag if } |I - \tilde{I}| > k\,\hat{\sigma}_R, \qquad \hat{\sigma}_R = 1.4826\,\text{MAD}, \qquad \text{MAD} = \text{median}_i|I_i - \tilde{I}|, \qquad k = 3.5 \text{ (default)} \tag{7}$$

Here $\tilde{I}$ is the median count rate. Both halves are chosen for resistance:

- The median center has a breakdown point of ≈50 %, so it is not dragged by the outliers being looked for — an arithmetic mean is, and a handful of dust spikes can shift it far enough that the *clean* samples fall outside the band.
- The factor 1.4826 makes $\hat{\sigma}_R$ consistent with σ for Gaussian data [8].

The same identifier recurs in the literature under different names, all consistent with the default above:

- Iglewicz and Hoaglin write it as a modified z-score, $M_i = 0.6745(I_i - \tilde{I})/\text{MAD}$ flagged at $|M_i| > D$, and recommend $D = 3.5$ from a simulation study; since $1/0.6745 = 1.4826$ the two forms are identical, and k here *is* their D.
- Davies and Gather give the same identifier its general treatment and prove its breakdown optimality [9]; the influence-function framework it rests on is Hampel's [10].
- Davies and Gather also report an asymptotic form of the rule, $|I - \tilde{I}| > 5.2\,\text{MAD}$ [9, p. 790], which they attribute to a later paper of Hampel's (1985, not held here); it agrees with the default above, since $3.5 \times 1.4826 = 5.19$.

When the scale collapses. If more than half the samples share one value — a flat, stalled or heavily quantized trace — the MAD is exactly zero and the band has no width. Pearson and co-workers call this an *implosion window* and characterize it for the windowed form of the same filter [11]; their treatment is to identify such windows and set them aside, and the rest of that paper's theory is stated for *implosion-free* sequences. This program does the same: it flags nothing and reports that it cannot judge, rather than marking every sample that is not exactly the median.

That has a cost, and it is deliberate — a single dust spike on a perfectly flat trace is not flagged either. The limit is inherent rather than a weakness of the MAD in particular. With over half the samples tied there is no scale to judge against, and the alternatives proposed precisely as improvements on the MAD, Rousseeuw and Croux's S_n and Q_n [8], carry the same 50 % implosion breakdown point and collapse on the same input.

On a shot-noise-limited trace the MAD scale is *not* the naive $\sqrt{\bar{I}}$ Poisson width, and the intuition that it should be is wrong. The MAD measures the trace's actual scatter, so at sub-second sampling the robust scale comes out *wider* than a $\bar{I} \pm k\sqrt{\bar{I}}$ band — about five-fold at 40 ms bins — because $\sqrt{\bar{I}}$ is the Poisson width only for raw counts binned at one second, the true rate width being $\sqrt{\bar{I}/\Delta t}$. A band built on $\sqrt{\bar{I}}$ is therefore too narrow for sub-second count rates and flags nearly every sample.

One further limit is worth knowing. The MAD of a Poisson sample can only land on the integer count lattice, so below roughly fifty counts per bin it becomes granular and estimates σ to within about a factor of two rather than a few percent — a very dim sample measured with short bins gets a coarse band.

The flagged fraction is a compact quality metric. The band is a *global* one: it flags departures from the trace's baseline of any duration without localizing them, so a slow drift trips it much as a single dust spike does. Localizing them is what the windowed companion below adds.

Windowed Hampel decomposition. The same identifier run against a *moving* baseline [11]. Over a window of $w = 2K + 1$ samples centered on each point k, take the median m_k and the robust scale S_k, and flag the center sample when it leaves the band:

$$m_k = \text{median}\{I_{k-K}, \ldots, I_{k+K}\}, \qquad S_k = 1.4826 \underset{j\in[-K,K]}{\text{median}} |I_{k-j} - m_k|, \qquad \text{flag if } |I_k - m_k| > k_\sigma S_k \tag{8}$$

where k_σ is the same band multiplier as Eq. 7 — one control drives both. Pearson's filter *replaces* a flagged sample with m_k; the program never does, reporting flags and the curves m_k and S_k instead.

The window sets what the layer can see. Pearson defines an impulse as a run of at most K samples, so an event shorter than about half the window can be flagged and anything longer is tracked as baseline — a persistent offset appears as a step in m_k rather than as flags. That blind spot is structural, not a defect, and showing m_k is what makes it visible.

The layer withholds a verdict in two cases, and reports the fraction of each. Where no full window fits — the first and last K samples — there is nothing to judge against; edge windows are deliberately not shrunk, since a short window is exactly the biased, implosion-prone regime the minimum width exists to exclude. And where more than K of a window's samples share one value the MAD is zero, Pearson's *implosion window*: no scale exists, so nothing is flagged there and none is substituted, exactly as for the global band above. A gap in the plotted band marks both.

Two notes on the width. Its default is resolved from the trace length, because the traces this program ingests span a seventy-fold range; the resolved value is displayed and exported. And no finite-sample MAD correction is applied: the minimum width is chosen so the uncorrected bias — $b_n = n/(n - 0.8)$ for $n > 9$ [12], Eq. 4, p. 414, about 7.8 % at the 11-point floor — stays below one step of the k_σ control, which keeps the regime needing a correction out of reach rather than correcting for it.

Fano factor. The count-rate histogram is fitted with a Gaussian and/or Poisson distribution; the key diagnostic is the Fano factor (variance/mean in count space):

$$F = \frac{\sigma^2}{\mu} \quad \begin{cases} \approx 1 & \text{shot-noise limited (ideal Poisson)} \\ > 1 & \text{excess fluctuations (slow modes, dust, drift)} \end{cases} \tag{9}$$

Blocking analysis. Cutting the trace into contiguous blocks of b samples and taking each block's mean gives a sequence $\bar{y}_k$ from which the standard error of the overall mean follows:

$$\text{SE}(b) = \frac{\text{std}(\bar{y}_k)}{\sqrt{m}}, \qquad m = \lfloor n/b \rfloor \tag{10}$$

For uncorrelated data this is flat in b. A curve that rises and plateaus is the signature of positive correlation — slow modes — and the plateau, not the $b = 1$ value, is the honest standard error.

Allan deviation. The same partition, read for *stability* rather than precision. Where the blocking curve asks how precise the mean is at a given block size, the Allan variance asks how reproducible consecutive blocks are, which is what says how long a trace may usefully be averaged. It is the mean squared difference of adjacent block means, halved [13], Eq. (11), p. 109:

$$\sigma_y^2(\tau) = \left\langle \frac{(\bar{y}_{k+1} - \bar{y}_k)^2}{2} \right\rangle, \qquad \hat{\sigma}_y^2(\tau) = \frac{1}{2m} \sum_{k=1}^{m} (\bar{y}_{k+1} - \bar{y}_k)^2 \tag{11}$$

with $\tau = b\,\Delta t$ the averaging time and m the number of differences averaged. The measure originates with Allan [14], which is why it carries his name; the compact difference form above is not printed there, and Barnes et al. are cited for it because theirs is the paper that states it and names it. The definition holds for contiguous gates — no dead time between measurements, $T = \tau$ — which a uniformly sampled trace satisfies exactly. The program reports σ_y in *fractional* form, dividing the block means by the trace mean, which is what makes the published slope mappings apply as printed [13], Eqs. (28)–(29), p. 111: white noise falls as $\tau^{-1/2}$, a random walk rises as $\tau^{+1/2}$, and a flicker floor is flat. The flat region is where averaging has stopped paying. One caution: a discrete spectral line — mains hum, a stirrer — makes $\sigma_y^2(\tau)$ vary rapidly with τ, and at a τ that happens to be a whole number of modulation periods the value can look far better than the trace deserves [13, p. 115]. Read this view beside the spectrum below, not alone.

Power spectral density. The diagnostics above describe how a trace's *amplitudes* are distributed; none of them can see whether anything in it is *periodic*. The program estimates the spectrum by Welch's method [15]: cut the record into overlapping segments of L points, window and transform each, and average the resulting periodograms.

$$I_k(f) = \frac{L}{U} |A_k(f)|^2, \qquad U = \frac{1}{L} \sum_{j=0}^{L-1} W^2(j), \qquad \hat{P}(f) = \frac{1}{K} \sum_{k=1}^{K} I_k(f),$$
$$S(f) = 2\,\Delta t\,\hat{P}(f) \quad [\mathrm{cps}^2/\mathrm{Hz}] \tag{12}$$

Mind the second line, which is not Welch's. He works throughout in units where the sampling interval is one and reports $\hat{P}$ alone; what the program plots and exports is a **one-sided spectral density in cps²/Hz**, and converting to it costs a factor $2\,\Delta t$ — the 2 for folding the negative frequencies onto the positive ones, the Δt for going from power per bin to power per hertz. Without it the numbers would be a factor $2\,\Delta t$ from the axis label, and — the reason it matters here rather than being pedantry — they could not be compared with the shot-noise floor two paragraphs below, which is the comparison this whole view exists to support.

Averaging is not a refinement here but the point: a single periodogram has 100 % relative scatter at every frequency however long the record, so an unaveraged spectrum cannot be read. How much scatter survives is set by the equivalent degrees of freedom of the approximating chi-square distribution, which depends on how strongly the overlapping segments are correlated [15, pp. 71–72]:

$$\rho(j) = \frac{\left[\sum_k W(k)W(k+jD)\right]^2}{\left[\sum_k W^2(k)\right]^2}, \qquad \text{E.D.F.} = \frac{2K}{1 + 2\sum_{j=1}^{K-1} \frac{K-j}{K}\,\rho(j)} \tag{13}$$

where D is the offset between segment starts. With no overlap ρ vanishes and this reduces to Welch's E.D.F. $= 2K$. The program computes it from the window actually in use, so the reported figure is right for a modern Hann window as well as for Welch's own two; it halves at zero frequency and at Nyquist. Segment length and window are user-owned — Welch treats the segment length purely as a resolution-versus-variance trade and prescribes no value. The overlap is fixed at half a segment, which is his own recommendation for a record of fixed length, and the program exposes no control to change it.

That number is what an error bar on the spectrum is built from. Welch names the approximating chi-square distribution and gives its degrees of freedom but prints no interval, so the band the program reports is the standard one constructed from that distribution — ours, not his:

$$\frac{\nu\,\hat{P}(f)}{\chi^2_{\nu,\,1-\alpha/2}} \;\le\; P(f) \;\le\; \frac{\nu\,\hat{P}(f)}{\chi^2_{\nu,\,\alpha/2}}, \qquad \nu = \text{E.D.F.} \tag{14}$$

The two multipliers are exported rather than the band itself, so a reader can apply them to whichever points they care about. Being a chi-square interval it is asymmetric — the upper reach is always the longer one — and it is a statement about the *estimator's* scatter alone: it says nothing about whether the detector was linear or the trace stationary.

The shot-noise floor. A spectrum needs a reference, and for photon counting there is a natural one. Counts in a bin of width Δt have variance equal to their mean, $\lambda = \bar{r}\Delta t$, so the count *rate* $r = n/\Delta t$ has variance $\bar{r}/\Delta t$; a white sequence of variance V sampled at $f_s = 1/\Delta t$ has one-sided spectral density $2V/f_s$. Hence

$$S_{\text{shot}} = 2\,\frac{\bar{r}}{\Delta t}\,\Delta t = 2\bar{r} \quad [\text{cps}^2/\text{Hz}] \tag{15}$$

flat at twice the mean count rate and independent of the sampling interval. This assumes dead-time-free Poisson detection; a detector with appreciable dead time is sub-Poissonian and sits below it. Broadband power above the floor is the same excess the Fano factor totals, resolved by frequency; a narrow peak above it is a candidate periodicity — whose significance is a separate question, answered next.

Periodicity significance. A spectrum always has a highest point, so "there is a peak" means nothing without asking how often noise alone would produce one that tall. For Gaussian noise the power at a given frequency is exponentially distributed *when measured in units of the data variance* σ_0^2, and the largest of N independent powers then has false-alarm probability [16], Eq. (14), p. 839

$$\Pr\{Z > z\} = 1 - \left[1 - e^{-z}\right]^N, \qquad Z = \max_n P(\omega_n), \qquad N = N_0/2 \tag{16}$$

with $N = N_0/2$ the independent frequencies of an evenly sampled record [16], Eq. (5), p. 836. Two details decide whether this number means anything. First, the null belongs to the **plain, unaveraged** periodogram normalized by σ_0^2 — Welch's averaged estimate is chi-square with the degrees of freedom above, not exponential, so the program runs the test on the periodogram and never draws this level against the spectrum. Second, the null assumes noise that is white as well as Gaussian. A drifting or correlated trace breaks that assumption in the direction that manufactures convincing low-frequency peaks, so when the blocking analysis reports correlations the verdict is shown **marked unreliable** rather than withheld or believed.

2.2 $\rho = R_g/R_h$

The shape parameter links the radius of gyration (R_g, from SLS) and the hydrodynamic radius (R_h, from DLS) of the same sample:

$$\rho = \frac{R_g}{R_h} \tag{17}$$

Both should be infinite-dilution values for ρ to be meaningful; the program checks that the two measurements share polymer, solvent, and temperature. ρ is a *qualitative* architecture fingerprint — the θ-solvent reference values below are from [17], Table 4 (primary derivation [18]), with the classical coil/sphere limits [1], [19]. Branched and star objects sit *below* the linear-coil value, but their ranges overlap the sphere-to-coil span, so ρ narrows the possibilities rather than naming one architecture.

ρ (θ-solvent)	Architecture
0.3–0.6	Microgel / internally cross-linked sphere
$\approx$ 0.78	Hard sphere (compact, near-spherical)
$\approx$ 0.98	Dendrimer / "soft sphere"
$\approx$ 1.08 $\rightarrow$ 1.33	Regular star — falls as arm number rises ($\approx$ 1.33 four-arm, $\approx$ 1.08 many-arm)
$\approx$ 1.23–1.25	Hyperbranched / cyclic
$\approx$ 1.50 (θ) – 1.78 (good solvent)	Linear random coil
$\approx$ 1.73	Broad (polydisperse) coil, or randomly branched
$\approx$ 2.05	Broad (polydisperse) coil in a **good** solvent — the upper coil range
$\geq$ 2.2	Extended / rigid-rod (ρ grows as ln N), or aggregated

Two caveats travel with these numbers [17]: **polydispersity raises** ρ (a θ-coil shifts 1.50 $\rightarrow$ 1.73 for a broad distribution), and the Kirkwood–Riseman theory behind the tabulated values underestimates the hydrodynamic interaction, so measured ρ tends to run ~10–14 % *below* them. They remain guides, not hard boundaries. The practical consequence — which the program states on the ρ reading itself — is that **a real sample can read one row too low**: a narrow θ-solvent coil, tabulated at 1.50, commonly measures 1.29–1.35 and so falls in the star/branched range. The two caveats work against each other, so the offset bites hardest on **narrow** samples, where no polydispersity shift is there to cancel it. It applies only to the rows the Kirkwood–Riseman treatment produces — the coil, branched/star and dendrimer values. The **hard sphere** is exact geometry ($R_g = \sqrt{3/5}\,R$, $R_h = R$, no hydrodynamic-interaction approximation) and the **microgel** range is empirical, so neither carries the offset and the program does not state it on them. The hard-sphere row does, however, name the overlap it sits under: the same offset puts a dendrimer (tabulated 0.98) near 0.84–0.88, which falls inside the hard-sphere band, so a reading there says so. ρ **is flagged apparent if *either* input is apparent** — a single-condition R_g or R_h is never silently combined with a thermodynamic one. For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

2.3 Scaling laws (R_g–M_w, A_2–M_w)

Across a homologous series (one polymer/solvent, varying molecular weight), size and the second virial coefficient follow power laws in M_w:

$$R_g = k_g\, M_w^{\nu} \tag{18}$$

$$A_2 = k_a\, M_w^{-a} \tag{19}$$

The program fits each as a straight line in log–log space; the **exponent** is the physically meaningful quantity and is independent of the (calibration-dependent) prefactor. Typical flexible-coil values: $\nu \approx 0.588$ in a good solvent, $1/2$ at the θ-point, $1/3$ for a collapsed globule or hard sphere; $a \approx 0.2$–0.3 [20], [21]. The fit needs at least two samples with the relevant pair. Below each plot the tab prints a **plain-language reading** of the fitted exponent (e.g. "$\nu \approx 3/5 \rightarrow$ swollen coil, good solvent"), mirroring the ρ table's architecture hint — a guide, not a hard boundary.

2.4 Branching contraction factor g

For flexible chains of the same chemistry and molar mass, a **branched** molecule is more compact than its linear analog. The **scattering branching contraction factor** compares their mean-square radii of gyration at matched M_w [22]:

$$g = \left.\frac{\langle R_g^2\rangle_{\text{branched}}}{\langle R_g^2\rangle_{\text{linear}}}\right|_{M_w} = \left(\frac{R_{g,\text{br}}}{R_{g,\text{lin}}}\right)^2 \le 1. \tag{20}$$

The platform reads the linear reference $R_{g,\text{lin}}$ from the **linear-only** R_g–M_w power law (Eq. 18, fit to the samples the user marks *linear*) evaluated at the branched sample's M_w. For a **regular f-arm star** the contraction factor and its inverse are [17], [22]:

$$g = \frac{3f-2}{f^2}, \qquad f = \frac{3+\sqrt{9-8g}}{2g} \quad (0 < g \le 1). \tag{21}$$

so a measured g returns an **effective arm number** f_{eff} — a *regular-star equivalent*, not a literal count (it assumes uniform arm length, and blurs for polydisperse arms and high f). g is a ratio of radii, hence **scale-independent** (it survives an uncalibrated run when the M_w scale is consistent across the samples). **Uncertainty is honest:** g and f_{eff} carry a $\pm$ only from **replicate** branched measurements — the replicate standard error of the mean (§6.4) — not a single-shot analytic SE. That $\pm$ reflects the **branched replicate spread only**; it treats the fitted linear reference law as exact and does **not** propagate the reference-law fit uncertainty. This is **guidance**, like the ρ labels: g itself is model-free; only the arm number assumes a star.

2.4.1 Random branching — the branching index n Real branched materials (radical/condensation polymers, hyperbranched, crosslinked gels) are rarely regular stars: their branch points fall at random, so a measured g maps not to an arm number but to a **branching density**. For a **randomly-branched fraction** — molecules of one molar mass with branch units

distributed at random — Zimm and Stockmayer give closed forms for f-functional branch units, with n the average number of branch units per molecule [22], Eqs. 47a/47b, [17], Eqs. 65/66:

$$g_3 = \left[\left(1+\tfrac{n}{7}\right)^{1/2} + \tfrac{4n}{9\pi}\right]^{-1/2}, \qquad g_4 = \left[\left(1+\tfrac{n}{6}\right)^{1/2} + \tfrac{4n}{3\pi}\right]^{-1/2}. \tag{22}$$

Each $g_{\mathrm{f}}(n)$ decreases monotonically from 1 ($n = 0$) toward 0, so the platform inverts it **numerically** for $n \geq 0$ (a bracketing root-find [23]) once the user selects a **tri-** or **tetra-functional** model. The headline caveat is **model-dependence**: the same g returns a *different* n under a different functionality (and a different f_{eff} as a regular star), so the reading always names the assumed model. These closed forms are the **θ-solvent** (ideal-chain) result and apply in good solvents too: because g is a *ratio* of branched-to-linear sizes at matched conditions, excluded-volume swelling **largely cancels** and g depends only weakly on solvent quality [24], [25]. Uncertainty follows the same rule as f_{eff}: n is a deterministic transform of the reported g, so it carries **no separate ±** — its uncertainty is g's (§6.4).

2.5 Hydrodynamic contraction factor *h*

The **hydrodynamic** sibling of g compares **hydrodynamic** radii instead of radii of gyration. Where g uses R_{g} (SLS), h uses R_{h} (DLS) [18]. It is defined through the translational diffusion coefficients (equivalently, the inverse-radius averages), so at matched M_{w}:

$$h = \left.\frac{D_{\mathrm{linear}}}{D_{\mathrm{branched}}}\right|_{M_w} = \frac{R_{h,\mathrm{br}}}{R_{h,\mathrm{lin}}} \leq 1. \tag{23}$$

The linear reference $R_{\mathrm{h,lin}}$ is read from the **linear-only** R_{h}–M_{w} power law (fit to the *linear*-marked samples) at the branched M_{w}. For a **regular *f*-arm star** the hydrodynamic factor follows its **own** relation — **not** g's $(3f-2)/f^2$ — the Kirkwood/Zimm pre-averaged, non-draining, θ result [18], Table II:

$$\begin{aligned} h &= \frac{\sqrt{f}}{(2-f)+\sqrt{2}\,(f-1)} = \frac{\sqrt{f}}{(\sqrt{2}-1)(f+\sqrt{2})}, \\ f &= \left(\frac{1+\sqrt{1-4\sqrt{2}\,a^2}}{2a}\right)^2, \; a = h(\sqrt{2}-1) \quad (0 < h \leq 1). \end{aligned} \tag{24}$$

Because the hydrodynamic contraction is **weaker** than the geometric one, a star gives **h > g** at every f (e.g. four-arm: $g = 0.625$, $h = 0.892$). Comparing g and h (or their ratio) is itself a topology / draining indicator. The program reads h against a **±5 % dead band** around unity, taken *multiplicatively* — $1/1.05 = 0.952$ to 1.05. Inside it, h is reported as little or no hydrodynamic contraction, *consistent with a linear (unbranched) chain*; only $h < 0.952$ is turned into an equivalent arm number through Eq. 24. The band is a reporting tolerance chosen to keep ordinary measurement scatter from being read as architecture, not a value from the literature.

It is one tolerance on a *radius*, applied to both factors in their own units:

- h is already a radius ratio, so its cut is 1.05.
- g is a **squared** radius ratio, so the same 5 % excess is $g = 1.05^2 \approx 1.10$.

A single number used on both would mean 5 % on a radius for h and only 2.5 % for g — tighter on the noisier of the two measurements. The band's deficit twin is the **inverse**, $1/1.05 = 0.952$, not $2 - 1.05$. Every comparison that mixes the two factors — including g vs h itself — is made in radius units for the same reason; [18] compares each factor to unity, which the square root leaves unchanged, and states no g-vs-h ordering, so nothing in the source argues for the raw pair. h is a ratio of radii, hence **scale-independent**, and its uncertainty is **honest** in the same way as g: a $\pm$ appears only from **replicate** branched measurements (a linear pass-through of the R_h replicate SEM, §6.4), never a single-shot analytic SE, and — like g — it reflects the branched replicate spread only, **excluding** the R_h–M_w reference-law fit uncertainty. g and h are **separate quantities with separate star relations** and are never conflated.

Above unity is not automatically an input error. For a **polydisperse** or **randomly-branched** architecture the z-average ratio can rise *above* 1 even though each molecule is more compact: the broad size distribution inflates $\langle R^2 \rangle_\mathrm{z}$ against a narrow linear reference, exactly as **polydispersity raises** ρ (§2.2). An A_f polycondensate — Flory/Stockmayer notation for a monomer carrying f identical reactive groups, condensed at random — reaches $g \approx 2$ and $h \approx 1.2$–1.4 over $f = 3 \rightarrow 6$ [18], Table II. The program therefore names that branched-architecture reading alongside the input-error one whenever g or h exceeds unity, while still excluding the **regular star**, which never rises above $h = 1$.

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3. Dynamic Light Scattering (DLS) Analysis

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. All methods fit $g_2(\tau)-1$ **directly** (β floating) and carry the Siegert factor of 2 explicitly, so a reported decay rate Γ is the physical g_1 rate; then $D = \Gamma/q^2$ and R_h follows from Stokes–Einstein, Eq. 2 (which needs the viscosity; decay rates and diffusion coefficients do not).

The two correlation-to-quantity conversions used below appear first as data forms when loading a generic file. For the `g2` form, the baseline is removed by

$$g_2(\tau) - 1 = \frac{G(\tau) - B}{B} \tag{25}$$

For the `g1` form, the **Siegert relation** [1] converts the field autocorrelation to the measured quantity:

$$g_2(\tau) - 1 = \beta \, |g_1(\tau)|^2 \tag{26}$$

where β is the instrument coherence factor (an optical property, typically 0.01–1.0 for standard goniometers), determined independently.

3.1 Parametric Methods

Cumulant expansion [26]**.** The standard first analysis. Taking the log of the baseline-subtracted signal and fitting a polynomial in τ:

$$\ln[g_2(\tau) - 1] = \ln \beta - 2\bar{\Gamma}\,\tau + \mu_2\,\tau^2 - \tfrac{1}{3}\,\mu_3\,\tau^3 + \cdots \tag{27}$$

The mean decay rate $\bar{\Gamma}$ gives $\bar{D} = \bar{\Gamma}/q^2$ and R_h; the second cumulant gives the **polydispersity index**

$$\mathrm{PDI} = \frac{\mu_2}{\bar{\Gamma}^2} \tag{28}$$

and the third (order-3 fit) the skewness. The cumulant result is reliable only up to PDI ≈ 0.3; the program flags whether this is satisfied. Above it, use a distribution method. Eq. 27 is the **linear** (log-polynomial) method; it assumes the baseline is exactly zero and is fit by weighted linear least squares.

Size range. PDI reports on the *fit*; it says nothing about whether the resulting size lies where DLS can measure at all. Sedimentation sets the upper bound — [27] puts it "much less than 10 µm", stated as a **diameter**, so the radius the program reports is compared against half that — and it flags any fitted

$$R_\mathrm{h} > R_\mathrm{h}^\mathrm{max}, \qquad R_\mathrm{h}^\mathrm{max} = 5\ \mu\mathrm{m} \quad \text{(engine default)} \tag{29}$$

The value is still reported, not withheld: a genuinely large scatterer is real data, and the flag says the number sits outside what the technique establishes rather than that the fit failed. The limit is the technique's, not the cumulant's, so every DLS method applies it — the single/double-exponential modes and both KWW sizes as well. The companion lower bound is the degeneracy floor at 0.2 nm, below which a fit has chased a short-lag artifact (detector afterpulsing in the first channels) rather than a particle. The nonlinear cumulant *rejects* such a solution, because a free floating-baseline fit that lands there has no measurement behind it; the linear cumulant and the exponential modes *flag* it, and the flag says what removes the artifact (skip the leading channels).

Did the correlogram decay? The size range is asked only of a decay the measurement observed. Before it, the cumulant fit asks the lag-window question of Eq. 34 in its **data** form — did the windowed correlogram ever fall to a fiftieth of its starting amplitude? — which is the form [27] itself states for the cumulant fit's retained range (Annex A.1.2). The rate form $\Gamma{\cdot}\Delta\tau$ is the wrong instrument here: on a broad or aggregating sample the mean rate is dominated by the fast component, so $\Gamma{\cdot}\Delta\tau$ reads many relaxation times "covered" while the correlogram plainly never decayed (measured on this program's aggregating test sample: the rate form flags none of its runs, the data form all of them, and no healthy run). A correlogram that fails it is flagged with that reason, and the size question is not asked of it at all — the size of an unobserved decay is not a size. Finally the polydispersity limit above is a precondition of the *size* too: a PDI past 0.3 flags the R_h, not only the PDI cell, so the verdict travels with the value into the Summary tables, the exports and the Cross-Sample candidates.

Nonlinear cumulant [28]**.** The program's **default** cumulant method fits $g_2 - 1$ *directly* by nonlinear least squares, expanding about the mean rate (Frisken's moments-about-the-mean form) with a **floating baseline** B:

$$g_2(\tau) - 1 = B + \beta\Big[\, e^{-\bar{\Gamma}\tau}\big(1 + \tfrac{1}{2}\mu_2\tau^2 - \tfrac{1}{6}\mu_3\tau^3 + \cdots\big)\Big]^2 \tag{30}$$

It yields the same $\bar{\Gamma}$, μ_2, PDI and R_h as the linear method but is more robust to baseline drift and noisy/low-count data, because B and the coherence β are fit rather than assumed and the whole windowed correlogram is used (no amplitude cutoff). Expanding about $\bar{\Gamma}$ (rather than about $\tau = 0$, as the log-polynomial does) keeps it stable at large τ. The baseline is constrained to small drift about the long-time tail so the fit cannot absorb the decay into B; if it fails to converge it reports no values at all rather than a substitute, and the result is labeled *(failed)*. The method is a **global setting** (**Settings → Cumulant method**, default *Nonlinear*) that applies to every cumulant-based step (per-measurement cumulant, Γ-vs-q^2, replicate averaging); changing it clears existing cumulant-based results so the workspace stays self-consistent. **Uncertainty:** like the linear method, the nonlinear fit reports **no ± from a single correlogram** (its lag channels are correlated [29], so the fit covariance under-reports); the honest DLS ± comes from replicate averaging (Eq. 101).

Analysis window: delay bounds + leading-channel skip. Every fit (cumulant, exponentials, KWW, NNLS/CONTIN/lognormal) selects which correlogram points it uses through one shared window with two complementary restrictions, applied in order:

1. a **leading-channel skip** — drop the first N correlator channels *by index* (the shortest, finest-spaced lags). The first few channels of a multi-τ correlator carry detector **afterpulsing** and dead-time artefacts that are not diffusion: a spuriously high first channel mimics a fast decay and biases the fit toward a falsely small R_h. The skip is an integer set in **Settings → Skip initial lag channels** (default 0 = keep all);
2. a physical **delay-time window** $[\tau_{min}, \tau_{max}]$ in time.

The two **compose as an intersection** — the effective first fitted point is the later of "channel N" and "first channel with $\tau \geq \tau_{min}$". The skip is applied **first**, so a corrupted leading channel cannot distort the amplitude cutoff/intercept (cumulant) or the β-to-$\tau\to0$ extrapolation (distributions). Use it sparingly with distribution methods: the short lags constrain the *fast / small* end of the size distribution, so an aggressive skip that a cumulant *mean* merely nudges can erase a genuine small-particle population entirely.

Single exponential — a monodisperse sample (equivalent to the first cumulant for clean data):

$$g_2(\tau) - 1 = \beta\, e^{-2\Gamma\tau} \tag{31}$$

Double exponential — two discrete modes through the full Siegert relation (appropriate when a polymer and a slow aggregate coexist); the fit reports whether it converged:

$$g_2(\tau) - 1 = \beta \left[\, f_1\, e^{-\Gamma_1\tau} + (1 - f_1)\, e^{-\Gamma_2\tau} \,\right]^2 \tag{32}$$

Kohlrausch–Williams–Watts (stretched exponential) [30] — broadly distributed or glassy dynamics, stretch exponent $s \in (0, 1]$ (s = 1 recovers a single exponential):

$$g_2(\tau) - 1 = \beta\, \exp[-2\,(\tau/\tau_c)^{\,s}]\,, \qquad \langle\tau\rangle = \frac{\tau_c}{s}\,\Gamma\!\left(\frac{1}{s}\right) \tag{33}$$

Both the characteristic time τ_c and the mean time $\langle\tau\rangle$ (via the gamma function) are reported, each with its hydrodynamic radius.

Did the mode actually decay? A parametric fit will happily report a rate for a mode whose decay the measurement never saw. The correlogram then constrains only the *start* of that decay, and the fitted Γ — with the R_h that follows from it — is an extrapolation past the end of the data rather than a measurement. It is a common condition, not an exotic one: a static or aggregated sample is the usual cause, and nothing else in the fit remarks on it, since the model fits the data it was given perfectly well.

The measurable question is how much of the decay the lag range covered. ISO's criterion has two equivalent expressions, and the program uses whichever the fit can answer:

$$\Gamma\,\Delta\tau < \tfrac{1}{2}\ln 50 \approx 1.96 \qquad \text{or} \qquad \frac{\min_j [G_2(\tau_j) - A]}{G_2(\tau_1) - A} > \tfrac{1}{50} \quad \text{(engine default)} \tag{34}$$

The left form is a **rate** test: for a mode of decay rate Γ measured across a lag span $\Delta\tau = \tau_\mathrm{max} - \tau_\mathrm{min}$, the dimensionless product $\Gamma{\cdot}\Delta\tau$ counts the relaxation times observed. The right form is the same requirement asked of the **data** — did the measured correlogram ever fall to a fiftieth of where it started?

The threshold is [27]'s, not an invention here. Annex A.1.2 fixes the range of points a cumulant fit may retain and requires at least one of them to fall below 1/50 of the initial amplitude — stating no consequence for a range that does not, so the program's choice to flag rather than discard departs from nothing; §10.2.1 a) states the same condition qualitatively, as a delay-time range that "covers the set of decay times". Since the measured amplitude $[G_2(\tau) - A]$ follows $|g_1|^2 = \mathrm{e}^{-2\Gamma\tau}$, the 1/50 requirement *is* $2\Gamma{\cdot}\Delta\tau \geq \ln 50$ — which is why the two forms above are one criterion. Xu [31] states the same quantity directly as a delay-range rule of thumb, $\tau_\mathrm{max} = 2.5/\Gamma$, and gives an independent, stricter version of the data form; on this program's test data the two select the same runs.

Which form applies, and why it matters. The rate form assumes the mode decays exponentially, so it is used where that is true and where the question is *per mode*: the single and double exponential fits, whose modes are judged independently. The stretched exponential (KWW) gets the **data** form instead, because its rate parameter cannot answer the question — a KWW absorbs a decay it never observed into its *stretch exponent* rather than into τ_c, so a τ_c-based test passes correlograms that plainly never decayed. Its two reported sizes share one verdict, since they are two readings of one decay. The **cumulant** fit takes the data form for the same reason (§3.1): its mean rate absorbs an undecayed slow component into the higher cumulants, and the Frisken form into its floating baseline as well. Note the data form takes the **minimum** over the fitted window, not the last point: a decayed correlogram's tail is noise about zero, and whichever channel lands last may sit back above the threshold. It also assumes the correlogram's baseline is already removed — a residual drift offset larger than the threshold reads as an incomplete decay.

What the standard states for a whole correlogram and one mean rate, this program applies per mode wherever the fit resolves modes — a two-mode fit routinely pairs a measured fast mode with an undecayed slow one, and a single verdict cannot say that. That per-mode step is this program's extension; the criterion and its threshold are the standard's.

The flag is per mode: the healthy mode beside an undecayed one keeps a clean number. This is deliberately **independent of the size range** of Eq. 29 — a mode can be larger than 5 µm and still have decayed, in which case a real, very large scatterer was genuinely measured and only the size range speaks. The two criteria answer different questions: one asks whether sedimentation has overtaken diffusion, the other whether the decay was observed at all — and the size question is

asked only when the answer to the second is yes, since the size of an unobserved decay is not a size. As with the size range, the value is reported rather than withheld, and the threshold is an adjustable engine parameter.

A multi-angle or multi-concentration fit inherits these verdicts: the Γ-vs-q^2 and D-vs-c lines (§3.3) carry a reliability flag of their own that is false when any per-point fit they rest on was flagged, or when the pooled R_h itself falls outside the size range, so a flagged single-angle value is never laundered into an unflagged pooled one.

3.2 Distribution Methods

These recover a full distribution of decay rates without assuming a functional form. The field autocorrelation is a weighted sum of exponentials,

$$g_2(\tau) - 1 = \beta \left[\int G(\Gamma)\, e^{-\Gamma\tau}\, \mathrm{d}\Gamma \right]^2, \tag{35}$$

discretized on a grid of hydrodynamic radii (default 1–1000 nm, log-spaced). We invert the **field ACF** g_1 directly [2, Sec. 8.11]: the single-Γ kernel is $A[m,n] = \exp(-\Gamma_n\, \tau_m)$ and the data vector is g_1 recovered from the Siegert relation, $\mathbf{y} = \mathrm{sign}(u)\sqrt{|u|}$ with $u = (g_2{-}1{-}\mathrm{baseline})/\beta$ — a **sign-preserving** square root, so long-delay noise stays zero-mean rather than rectifying into a positive pedestal that a non-negative fit would read as spurious large-R_h weight. The reconstruction back to measured space is $g_2{-}1 = \beta(\mathbf{Ax})^2 + \mathrm{baseline}$. Both methods need β and a baseline (estimated automatically — β by extrapolating to zero delay, the baseline from the long-delay tail — and overridable; the values used are always reported).

The residual is **statistically weighted** by the delta-method inverse variance of the recovered g_1: propagating a (locally uniform) $g_2{-}1$ noise through the square root gives $\mathrm{Var}(g_1) \propto 1/|u|$, so the per-lag weight is $w \propto {g_1}^2$ ($\mathbf{M} = \mathrm{diag}(w)$), down-weighting the noise-dominated long-delay channels. The weight's overall scale cancels in the fit at any **fixed** value of the regularization parameter α (introduced below) — and everywhere in NNLS — so it needs no user input. It does **not** cancel in the automatic α selection: rescaling w by c reparametrizes Eq. 36 as $\alpha \to \alpha/\sqrt{c}$, so the selected α slides by $\sqrt{c}$ across the fixed sweep — and the per-run β enters the weight as exactly such a rescale ($w \propto |u| = |g_2{-}1{-}\mathrm{baseline}|/\beta$). The selectors re-minimize and track the shift, leaving the recovered distribution nearly unchanged unless the tracked optimum reaches an end of the sweep (a ceiling pick is flagged on the result; a floor pick is legitimate on clean data and is not); a user-supplied fixed α does not compensate. The weight is a **fit-only precision map — no statistical ± is reported** from it: the distributions stay uncertainty-free (a single correlogram's lag channels are correlated [29], and the off-diagonal covariance is large, so an honest ± is deferred to replicate averaging — §6.5).

- **NNLS** solves $\min\|\mathbf{M}^{½}(\mathbf{Ax} - \mathbf{y})\|^2$ with $\mathbf{x} \geq 0$ and no smoothing. It resolves well-separated peaks but produces spiky, noise-sensitive distributions.
- **CONTIN** [32] adds Tikhonov regularization [33]:

$$\min_{\mathbf{x}\geq 0} \|\mathbf{M}^{1/2}(\mathbf{Ax} - \mathbf{y})\|^2 + \alpha^2 \|\mathbf{Lx}\|^2 \tag{36}$$

with $\mathbf{L}$ the second-difference operator. The constrained problem is solved as an **augmented non-negative least-squares** stack — αL is stacked beneath the weighted design and the whole system solved with NNLS, so non-negativity and smoothing are imposed together (rather than a closed-form Tikhonov solve). The regularization strength α is chosen automatically by one of **three** selectors over the same sweep (the full sweep is returned and a fixed α can be supplied); each records on the result which selector chose it. CONTIN is generally better for broad distributions and real data.

GCV (generalized cross-validation) (default; [34]). GCV estimates the leave-one-out prediction error without knowing the noise level. For our weighted Tikhonov fit, with $\mathrm{tr}(\mathbf{I} - \mathbf{A}_\alpha) = N_y - \mathrm{NDF}(\alpha)$,

$$\mathrm{GCV}(\alpha) = \frac{N_y\, V(\alpha)}{\left(N_y - \mathrm{NDF}(\alpha)\right)^2}, \qquad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|^2, \tag{37}$$

where $V(\alpha)$ is the weighted residual, $\mathrm{NDF}(\alpha)$ the Tikhonov effective degrees of freedom (Eq. 40), and N_y the number of lag channels; the chosen α is the minimizer. GCV is σ^2-free and asymptotically optimal, and is the robust default for this ill-conditioned g_1 kernel — it resolves noisy multimodals the L-curve over-smooths. Its one documented failure mode is a **flat** minimum [35]:

- A minimum pinned to the **high-α** end of the sweep signals over-regularization and raises a visible caution (widen the range, or inspect the L-curve).
- A pick pinned to the **low-α** end raises the twin caution: the selector wanted a smaller α than the range allows. At a low floor, such as the default 1e-4, the pick is then essentially the unregularized NNLS fit, and extra peaks in it may be inversion noise rather than populations — the "solutions with extra peaks" of an unregularized inversion [36, Sec. 3.2.3]; at a user-raised floor it is simply the least-smoothed answer the range permitted.

CONTIN itself sweeps α relative to the problem's largest singular value, from machine precision up to that value [36, Sec. 4.1.3]; the fixed sweep here is a convenience simplification of that grid. Reading the manual's selection criterion (its Eqs 1–2, §4.1.6.5), the smallest-α solution is the F-test's *reference*, so the original program is unlikely to choose it — but the F-test implemented here measurably does pin to the floor on fine R_h grids, which is why both edges are flagged rather than trusted.

L-curve corner ([35]; corner-finding algorithm [37]). Plot the trade-off curve $(\log\|\mathbf{A}\mathbf{x}_\alpha - \mathbf{y}\|,\ \log\|\mathbf{L}\mathbf{x}_\alpha\|)$ and pick the α of **maximum curvature** — the corner of the "L". A cubic spline is fitted through the discrete points and its analytic curvature maximised:

$$\kappa(\alpha) = \frac{\zeta'\eta'' - \zeta''\eta'}{\left((\zeta')^2 + (\eta')^2\right)^{3/2}}, \qquad \zeta = \log\|\mathbf{A}\mathbf{x}_\alpha - \mathbf{y}\|, \quad \eta = \log\|\mathbf{L}\mathbf{x}_\alpha\|, \tag{38}$$

primes denoting derivatives along the curve, picking the α of maximum curvature. On this kernel it smooths more heavily than GCV, so it is offered as an alternative.

Provencher F-test ("probability to reject"; [33]; [38]). Among the sweep of increasingly-smoothed solutions, each is compared to the least-squares reference by the F-statistic

$$F(\alpha) = \frac{V(\alpha) - V_0}{V_0} \frac{N_y - \mathrm{NDF}_0}{\mathrm{NDF}_0}, \qquad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|^2, \tag{39}$$

where V_0 is the **minimum** residual over the sweep — at the reference α_0, the least-regularized (least-squares) end — N_y the number of data points, and $\mathrm{NDF}_0 = \mathrm{NDF}(\alpha_0)$ the **effective degrees of freedom at that reference**. Provencher's NDF (his Eqs. 3.15–3.16), $\mathrm{NDF}(\alpha) = \sum_j s_j^2/(s_j^2 + \alpha^2)$ with s_j the singular values, is exactly the trace of the Tikhonov influence ("hat") matrix [35] — *not* the raw grid size:

$$\mathrm{NDF}(\alpha) = \mathrm{tr}\big[\mathbf{A}(\mathbf{A}^\top\mathbf{A} + \alpha^2\mathbf{L}^\top\mathbf{L})^{-1}\mathbf{A}^\top\big] = \sum_j \frac{s_j^2}{s_j^2 + \alpha^2}. \tag{40}$$

Crucially NDF_0 is a **single fixed value** (evaluated at α_0), used in both the scaling factor and the F-distribution — not the per-α DOF. For an ill-posed Laplace kernel NDF is small (typically a handful to a couple dozen for a correlogram). $F(\alpha)$ is F-distributed with (NDF_0, $N_y - \mathrm{NDF}_0$) degrees of freedom, so its cumulative value $f_c(\alpha) = \mathrm{CDF}_F(F(\alpha);\, \mathrm{NDF}_0,\, N_y - \mathrm{NDF}_0)$ is the "probability to reject": ~0 where the regularizer barely changes the fit (under-smoothed) and ~1 where it changes it a lot. The chosen α is the one whose f_c is closest to a user level (default 0.5); a **higher** level tolerates more fit degradation and selects a **smoother**, more-parsimonious distribution, a lower level a rougher one. The F-test treats the N_y residuals as independent, but a single correlogram's lag channels are correlated [29], so N_y overstates the independent information and the level is a **guide, not an exact test**. It is offered for comparison with legacy CONTIN output. No single criterion is best in every case, so all three selectors are user-selectable and their sweeps are returned for inspection [35].

- **Lognormal** is a *single-mode parametric* distribution [39]: it assumes the intensity-weighted size distribution is lognormal in R_h and fits its median and log-width to the same $\mathbf{A}\,\mathbf{x} = \mathbf{y}$ model. On the log-spaced grid the discrete weight is a house adaptation of the standardized lognormal density,

$$w_i \propto \exp\left[-\frac{(\ln R_{h,i} - \mu)^2}{2\sigma^2}\right], \qquad \mu = \ln(\text{median } R_h) \tag{41}$$

with σ the log-width (polydispersity); the $1/R_h$ of the lognormal pdf cancels the log-grid spacing, so the discrete weight carries no extra Jacobian. It cannot resolve multiple modes, but it is robust to noise and always returns a smooth, single-peaked distribution — a good default for a known-unimodal sample.

> **Caution (all distribution methods).** For the multi-modal inversions (NNLS and CONTIN), fine structure — small secondary peaks especially — is sensitive to noise and the regularization choice and is not always reproducible. Treat the main peaks as reliable; cross-check width against the cumulant PDI and the parametric fits. (Lognormal sidesteps this by assuming a single peak, at the cost of not resolving more than one.)

3.3 Combining Measurements

Γ versus q^2 (multiple angles). For pure translational diffusion the decay rate is proportional to q^2:

$$\bar{\Gamma} = D\,q^2 \tag{42}$$

A straight line through the origin gives D (and R_h). Three linearity diagnostics test whether the motion is genuinely diffusive: the fit R^2; the intercept of an unconstrained fit (should be ≈ 0 — a nonzero value indicates a non-diffusive contribution); and the trend of $D_\mathrm{app} = \Gamma/q^2$ with q^2 (should be flat — an upturn signals internal/segmental modes). A combined diffusive / non-diffusive flag summarizes these.

Concentration extrapolation. Extrapolating the apparent diffusion coefficient to zero concentration removes inter-particle interactions:

$$D_\mathrm{app}(c) = D_0\,(1 + k_D\,c) \tag{43}$$

The intercept D_0 gives the infinite-dilution $R_{\mathrm{h},0}$; the slope gives the diffusion interaction parameter k_D (positive in good solvents, negative in poor ones).

A concentration whose fit returns no usable decay rate — a non-positive Γ, which a weak scatterer on a noisy baseline can produce — is **dropped from the line** and named in a notice beside the result; the remaining concentrations are extrapolated normally. Its concentration is marked with a dotted vertical line on the D vs c plot and the drop is stated in the exported file's Comments cell, so a removed run is visible in every place the result travels rather than quietly absent. If fewer than two concentrations survive there is nothing to extrapolate, and D_0, $R_{\mathrm{h},0}$ and k_D are all withheld.

> Both refuse a set that **mixes polarizations**, naming the tags found. A sample may hold both VV and VH runs — that pairing is what DDLS is for (§3.4) — but $\Gamma_\mathrm{VH} = \Gamma_\mathrm{VV} + 6D_\mathrm{r}$ is a rotational term, not scatter, so a line through both has the wrong slope and a non-zero intercept that is not physical (run each from the **DLS tab**; User-Manual §10.1). Sets whose runs share one tag are unaffected.

Mode-resolved dynamics (the mode map). When a sample relaxes on more than one time scale — the fast + slow two-mode structure of salt-free polyelectrolytes [40], or the slow modes of neutral semidilute θ-solutions [41] — a single pooled Γ per angle averages physically distinct processes. The **Mode Map** (DLS tab) instead plots *every* extracted mode of *every* run in (q^2, Γ) space — the two-mode double-exponential's modes, or a CONTIN/NNLS distribution's resolved peaks — and the *user* picks the point sets that form mode families; the program deliberately performs **no automatic association**, because rank-based pairing breaks exactly where the analysis is interesting: where a diffusive mode's Γ sweeps past a q-independent mode's across the angle range. Each picked family is then fitted separately with Eq. (42) (through-origin D, the same linearity diagnostics), and with the slow-mode deviation form of Sedlák & Amis [40]:

$$\tau_s^{-1} = a\,q^2 + b\,q^4 \tag{44}$$

whose q^4 term captures the apparent (non-diffusive) character of a slow mode; a is the mode's $q \to 0$ apparent diffusion coefficient. (The deviation is the angular fingerprint Burchard's first-cumulant

expansion $\Gamma/q^2 = D_z(1 + C\langle S^2\rangle_z q^2 - \ldots)$ assigns to internal structure; his Table 4 puts C at 13/75 ≈ 0.17 for a monodisperse coil, 1/5 at $M_w/M_n = 2$, down to 2/15 ≈ 0.13 in its branched/star limits [42].)

Every family's R_h is computed by Stokes–Einstein (Eq. 2) and reported flagged **apparent and mode-resolved**, carrying several further diagnostics:

- For a slow mode that reflects collective or structural relaxation rather than a particle, Stokes–Einstein is not strictly applicable [43], so the number is an apparent size for comparison, never gated away.
- Whether the family passes through the origin is a **label carrying its stated tolerances** ($R^2 \geq 0.98$, |intercept|/mean $\Gamma \leq 0.1$ by default), never an eligibility test.
- A mode carrying less than ~2 % of the run's amplitude is flagged imprecise — the practical resolvability floor of the decomposition [44].
- The fast mode of a salt-free polyelectrolyte is the coupled polyion–counterion mode (the classic treatment: [2] §13.7; a modern account: [45] §7.6–7.7), whose amplitude is expected q-flat.
- The map's **amplitude report** plots A_f/A_s against $\sin^2(\theta/2)$ — the axis of Sedlák & Amis's Fig. 4 [40] — for the two role-designated families, with an angle-flatness diagnostic whose relative-span threshold (default 20 %) is a **house convenience parameter**: no printed criterion exists in the held sources, so it gates a note, never a verdict a result depends on.

Uncertainty: a mode's amplitude and its per-run Γ come from a **single correlogram and carry no ±** (the lag channels of one correlogram are correlated, so a regression SE would under-report — see §6); the per-family fits combine independent angles, so D, R_h, a and b carry ordinary regression SEs (statistical only). Replicate averaging remains the honest route to a single-sample dynamic ±.

3.4 Depolarized DLS (DDLS): rotational diffusion

For an **optically anisotropic** scatterer (a rigid rod, or an anisotropic particle), the **depolarized** (VH) correlogram decays faster than the polarized (VV) one because molecular *reorientation* adds to the decay. In the small-qL regime the two field (g_1) decay rates are

$$\Gamma_{VV} = q^2 D_t, \qquad \Gamma_{VH} = q^2 D_t + 6D_r, \tag{45}$$

where D_t is the translational and D_r the **rotational** diffusion coefficient [2], [46] (the dynamic-scattering-from-rotation framework originates with [47], which gives the translational and rotational broadening as separate limits). Subtracting isolates the rotation, and its inverse is the reorientation time:

$$D_r = \frac{\Gamma_{VH} - \Gamma_{VV}}{6}, \qquad \tau_{rot} = \frac{1}{6D_r}. \tag{46}$$

Each correlogram is fitted for its Γ with the cumulant engine (§3.1); since a homodyne *intensity* correlogram decays at $\mathbf{2}\Gamma$, the field rate Γ is what enters Eqs. 45, 46 — the factor of 2 is handled by the fitter, not double-counted here. Across angles, D_t is taken from the **VV** channel (Γ_{VV} vs q^2 through the origin — the whole VV signal, cleanly measured), and D_r from the mean of the per-angle differences. The $6D_r$ term is q-independent, so a plot of Γ_{VH} vs q^2 is a line of slope D_t and intercept $6D_r$ — the translational/rotational separation at a glance.

Validity. Eq. 45 is a single exponential only for $\mathbf{qL \lesssim 3}$ (above it, higher rotational modes and intramolecular interference appear; [46]). If the rod length L is supplied, qL is evaluated per angle and the result is flagged when it exceeds the limit.

Uncertainty. A single angle (one VV + one VH correlogram) carries **no** ± (one correlogram per channel; [29]). Across angles the points are independent, so D_t gets the through-origin regression SE and D_r the spread (SD/$\sqrt{n}$) of the per-angle estimates — both labeled statistical-only (§6). On a single angle the exported ± columns say *why* they are blank rather than keeping the multi-angle label, which would assert a statistical SE that was never computed.

When $\Gamma_{VH} \leq \Gamma_{VV}$. The difference in Eq. 45 is then zero or negative, so D_r is unphysical for an anisotropic scatterer — check the VV/VH pairing, the signal-to-noise, or whether the particle is optically isotropic. The verdict is a per-angle one, so it fires even when the *mean* D_r is positive, and it travels into the exported file: every column whose value is a *function* of D_r repeats it in that column's Comments cell — the per-angle values, the mean, its ±, $\tau_{rot} = 1/(6D_r)$, and every shape-model dimension of §3.5, each of which is inverted from the (D_t, D_r) pair. The shape models are the reason the per-angle test matters: a single bad angle can leave the mean positive, so the models still run and still report a length.

The D_t family is deliberately left unmarked, since D_t is the through-origin slope of Γ_{VV} vs q^2 and is unaffected by VH noise or by an optically isotropic scatterer. That is not a claim of immunity: a **swapped VV/VH pairing** corrupts both channels at once, which is why the note names that cause first — the split marks the number in doubt for the causes a reader can act on.

Practical note. The VH signal is intrinsically **weak** (the matrix/solvent scatters polarized), so depolarized acquisition needs a clean analyzer and long counting; flexible polymer coils barely depolarize, so DDLS targets rods and anisotropic particles. Mapping D_t, D_r to a rod's length and diameter (Tirado–García de la Torre) is described in §3.5.

> See the **User-Manual** §10.1 for how to run DDLS. Tag each correlogram's polarization (VV/VH) in the **Data tab** (the *Polarization* row); the sub-tab pairs a VV with a VH at each angle automatically. **Replicates** — several VV (or VH) correlograms at the *same* angle — are averaged (their fitted Γ values are meaned) rather than one being dropped, and are held to the same repeat-set test the sidebar averaging actions apply (§6.4): runs that differ in concentration or M_w fraction are refused, naming the channel and angle they were found in, rather than pooled. For weak VH, pre-averaging the correlograms themselves via the sidebar *Average correlation functions* action (then tagging the result) gives better signal-to-noise.

3.5 DDLS shape models: dimensions from D_t and D_r

D_t and D_r are **shape-free observables**; turning them into particle *dimensions* requires assuming a geometry, so the outputs below are **model-dependent** — dimensions of an *assumed* shape, never a direct measurement. Three models are offered, and read together they diagnose the shape.

Sphere (one unknown). The rotational diffusion of a sphere of radius R is the Stokes–Einstein–Debye relation [48], whose inverse gives R:

$$D_r = \frac{k_B T}{8\pi\eta R^3}, \qquad R = \left(\frac{k_B T}{8\pi\eta D_r}\right)^{1/3}. \tag{47}$$

For a *true* sphere this equals the translational (Stokes) radius $R_\mathrm{h} = k_\mathrm{B}T/(6\pi\eta D_\mathrm{t})$, so the program reports both and their ratio $R(D_\mathrm{r})/R_\mathrm{h}$: ≈ 1 means a sphere is plausible; a ratio above 1 is the **rod signature** (rotation reports a smaller effective size than translation), flagging that the sphere model does not apply.

Rigid rod (two unknowns). For a cylinder of length L and diameter d (axial ratio $p = L/d$), the Tirado–García de la Torre [49] relations are

$$D_t = \frac{k_B T}{3\pi\eta L}\,(\ln p + \nu), \qquad D_r = \frac{3k_B T}{\pi\eta L^3}\,(\ln p + \delta_\perp), \tag{48}$$

$$\nu = 0.312 + \tfrac{0.565}{p} - \tfrac{0.100}{p^2}, \qquad \delta_\perp = -0.662 + \tfrac{0.917}{p} - \tfrac{0.050}{p^2}. \tag{49}$$

With two observables and two unknowns the rod is exactly determined. There is no closed-form inverse, so the program solves it numerically: at a trial p, D_t fixes L (Eq. 48, left); the predicted D_r is monotonic in p, so a one-dimensional root find recovers p, then L and $d = L/p$.

Validity. The end-effect polynomials (Eq. 49) are fitted only for $\mathbf{2 < p < 30}$; a recovered p outside that range is flagged as an extrapolation, and if no rod can reproduce both coefficients the rod model is reported as not fitting.

Ellipsoid of revolution (two unknowns). For a prolate or oblate ellipsoid with polar semi-axis a and equatorial radius b (axial ratio $p = a/b$, so $p > 1$ is prolate, $p < 1$ oblate), both friction factors follow from Perrin's shape integral. In Perrin's variable $\rho = b/a = 1/p$, the dimensionless factor is

$$aS = \begin{cases} 2\,(1-\rho^2)^{-1/2}\,\ln\dfrac{1+\sqrt{1-\rho^2}}{\rho}, & \rho < 1 \text{ (prolate)},\\[2ex] 2\,(\rho^2-1)^{-1/2}\,\tan^{-1}\sqrt{\rho^2-1}, & \rho > 1 \text{ (oblate)}. \end{cases} \tag{50}$$

The **translational** coefficient is Perrin's friction factor as tabulated by Tanford [50]; the **rotational** coefficient — the end-over-end tumbling of the symmetry axis, which is what DDLS observes — is Koenig's corrected form [51] of Perrin's [52] expression:

$$D_t = \frac{k_B T}{6\pi\eta R_0}\cdot\frac{\rho^{2/3} aS}{2}, \qquad R_0 = (ab^2)^{1/3}; \tag{51}$$

$$D_\perp = \frac{k_B T}{C_2}, \qquad C_2 = \frac{4C_0\,(1-\rho^4)}{3\rho^2\,[\,aS\,(2-\rho^2)-2\,]}, \qquad C_0 = 8\pi\eta\,ab^2. \tag{52}$$

Both reduce to the sphere limits (Eq. 47 and Stokes–Einstein) as $p \to 1$. As with the rod, the 2-D inverse is reduced to a 1-D root find in p (at a trial p, D_t fixes the size, then $D_\perp$ is matched). Unlike the rod, an ellipsoid inverse is **not single-valued**: a given (D_t, $D_\perp$) can be reproduced by both a prolate and an oblate ellipsoid, and the oblate branch is non-monotonic (a maximum near $p \approx 0.72$), so up to **three** ellipsoids fit. The program reports **every** solution that fits — none is silently preferred — and flags the result as degenerate when more than one does.

Validity. Koenig's factors are tabulated for $\mathbf{0.1 < p < 10}$; a recovered p outside that band is flagged as an extrapolation.

Uncertainty. The sphere radius SE follows from D_r by the delta method ($R \propto D_r^{-1/3} \Rightarrow \mathrm{SE}(R)/R = 1/3 \cdot \mathrm{SE}(D_r)/D_r$). This is a single, smooth, monotonic power, so the delta method is accurate at realistic precision and degrades only for large $\mathrm{SE}(D_r)/D_r$; it is **Monte-Carlo-validated** (§6.5) — sampling D_r and re-inverting matches the delta SE to ~1.5 % at a stressing 10 % spread. The rod and ellipsoid inverses are nonlinear with no closed form, so their dimension SEs are propagated by **Monte Carlo** directly — sampling D_t, D_r from their SEs and inverting each (for the ellipsoid, each draw's solutions are matched to the base solution they belong to) — which is the sampling spread, with no linearization to validate. The inversion is solved over the model's whole domain (every $p > 1$ for the cylinder, every $p > 0$ for the ellipsoid), so a draw is lost only where no such shape reproduces it — a negative rate, or a D_r faster than the $p \to 1$ limit admits for that D_t — and not to a shape window; the remaining bracket ends are float limits placed far outside any particle light scattering sizes. The census (draws reproduced of draws taken, and the seed) is reported beside every ±, because the ± is the spread over exactly those draws. The aspect ratio's distribution is heavy-tailed at large input uncertainty (a slightly slow D_r draw is a much longer, thinner rod), so its ± can be large next to the point estimate; L and d stay well behaved. The rod ± is checked against the sampling SD of independent noise realisations (§6.5).

Mapping diffusion to dimensions is the heart of why DDLS is run on rods and anisotropic particles rather than flexible coils [48].

> Run from the same **DLS tab** → **DDLS** sub-tab: both models compute automatically after you press **Run** (the sphere needs the solvent viscosity).

3.6 Weighted size distributions (volume / surface / number)

The distribution methods of §3.2 return an **intensity-weighted** size distribution $w_I(R)$: each bin's weight is proportional to how much light that size scatters. Because large particles scatter far more than small ones, the intensity view over-represents them. Re-expressing the distribution as **volume-, surface-, or number-weighted** requires an optical model to convert "how much this bin scatters" into "how many particles are there." Writing $N(R)$ for the number distribution and $I_1(R,\, q)$ for the scattering per particle, the intensity weight of a bin is

$$w_I(R_i) \propto N(R_i)\, I_1(R_i, q) \;\Rightarrow\; N(R_i) \propto \frac{w_I(R_i)}{I_1(R_i, q)}, \tag{53}$$

and the other weightings follow from the geometric moments of a sphere,

$$w_{\mathrm{surface}}(R_i) \propto N(R_i)\, R_i^2, \qquad w_{\mathrm{volume/mass}}(R_i) \propto N(R_i)\, R_i^3. \tag{54}$$

This inversion is well-posed **only for homogeneous spheres**, where the DLS-measured hydrodynamic radius equals the geometric radius ($R_h = R$) and I_1 is a clean law; the reweighting therefore offers **sphere models only** and refuses a non-spherical choice. The scattering per particle depends on the size regime:

Rayleigh (point scatterer), valid for very small particles, $x = 2\pi R/\lambda \lesssim 0.1$ ($\lambda = \lambda_0/n$ the in-medium wavelength):

$$I_1(R, q) \propto R^6. \tag{55}$$

Rayleigh–Debye–Gans (RDG) sphere, valid at low contrast, $|m - 1| \ll 1$ and $2x|m - 1| \ll 1$, with $m = n_{\text{solute}}/n_{\text{medium}}$ the relative refractive index [53]:

$$I_1(R, q) \propto V^2\, P(qR) \propto R^6\, P(qR), \qquad V = \tfrac{4}{3}\pi R^3, \tag{56}$$

with P the sphere form factor (Eq. 76). RDG reduces to the Rayleigh R^6 law as $qR \to 0$. The bulk optical-contrast factor — proportional to $(m^2 - 1)^2$ ($\approx (2(m - 1))^2$ in the RDG small-contrast limit) — is the **same for every size bin** (one material), so it **cancels** in Eq. 53 after normalization: m enters only the RDG **validity gate**, never the reweighting arithmetic itself.

Full Mie sphere, the *general* case (any x, any contrast — §4.9): the per-particle intensity is the full Mie result $I_1 \propto |S_1(x, m, \theta)|^2$ (Eq. 82). Unlike RDG, the contrast does **not** cancel size-independently, so each bin carries its own x, m dependence; m (from the solute refractive index) is required. Mie reduces to the RDG law as $m \to 1$ and to Rayleigh as $x \to 0$, so it is the top of the **Rayleigh → RDG → Mie** ladder. It is what makes the classic high-contrast case (polystyrene latex in water, $m \approx 1.19$) work where RDG and Rayleigh must refuse.

Validity gating. The size parameter $x = 2\pi R/\lambda$, the contrast $|m - 1|$, and the product qR are checked per bin against documented default bounds (Rayleigh $x \leq 0.1$; RDG $|m - 1| \leq 0.1$, $2x|m - 1| \leq 0.1$, and qR below the sphere's first form-factor minimum 4.4934) [53]. The bound "$\ll 1$" is rendered as ≤ 0.1 following Bohren & Huffman (Ch. 6, p. 464: RDG errors grow as $|m - 1| \to$ ~0.1). **Mie has no upper contrast/size validity bound** (it is the general case), but a bin near a Mie form-factor minimum — where its normalized angular intensity $P(\theta) = |S_1|^2/|S_1(0)|^2$ falls below a documented floor (default 0.01) — is flagged like the RDG first-minimum gate, because $N = w_{\text{I}}/I_1$ spikes on noise there; a genuinely small particle ($P \approx 1$) is not flagged. Mie also carries a documented large-x convergence ceiling and an informational note where the cheaper RDG limit already agrees. Bins outside a model's range are **flagged** (a validity overlay marks them; the number/volume view is still drawn) — never silently reweighted with the wrong law — and the model can be set to **refuse** instead.

Uncertainty. Reweighted distributions carry **no** ±. They derive from the ill-posed NNLS/CONTIN inversion (which itself reports no SE, §6.5), and the $/I_1$ division — dominated by $/R^6$ — **amplifies** small-R noise into large spurious small-particle populations. They are report-only and model-dependent, distinct from the measured intensity distribution they derive from. This is why ISO 22412:2017 §9.1 recommends against — and specifically deprecates — number distributions from DLS [27]; the capability is provided (commercial software offers it) but honestly labeled and never presented as a measurement.

> See the **User-Manual** §10.1 for how to run this. Pick a **Weighting** (Intensity / Volume / Surface / Number) and, for a non-intensity view, a **Sphere model** (Rayleigh point / RDG sphere / Mie sphere). RDG and Mie read the hand-entered **solute refractive index** from the Data tab. Switching weighting re-presents the cached fit — it does not re-run it.

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4. Static Light Scattering (SLS) Analysis

The SLS module extracts the weight-average molecular weight M_w, the radius of gyration R_g, and the second virial coefficient A_2. It first converts raw intensities to absolute excess Rayleigh ratios, then constructs the Zimm, Berry, or Debye plots.

4.1 Calibration to an Excess Rayleigh Ratio

Raw intensities are placed on an absolute scale in steps: subtract the detector dark count, subtract the pure-solvent scattering (to isolate the excess from the polymer), correct for the scattering volume via $\sin\theta$, apply the refractive-index correction Eq. 3, [3], and scale by a calibration constant k_c:

$$\Delta R_\theta = k_c \, \sin\theta \, (I_{\text{sample}} - I_{\text{solvent}}) \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}} \right)^2 \tag{57}$$

The program computes its own k_c from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's Rayleigh ratio (§1.5, in the matching geometry) [1]:

$$k_c = \frac{R_{\text{std}}}{I_{\text{cal,net}} \, \sin\theta_{\text{cal}}} \tag{58}$$

The constant is shown and can be edited. Any value from a data file is kept only for reference and not used in the calculation — a stored k_c goes stale after a re-alignment, and vendor Rayleigh ratios are often outdated.

> **Uncalibrated fallback.** If no k_c can be computed or supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated. Then M_w and the absolute A_2 are marked unreliable — but R_g is unaffected (it comes from the ratio of slope to intercept, where the scale cancels), and so is the calibration-free product $2A_2M_w$ (§4.4). Those remain valid without calibration.

4.1.1 Derived Intensities I_{sample} in Eq. (57) need not come from a static scan. The scattered intensity a static experiment measures *is* the time average of the photon count rate, so a run recorded for DLS supplies it directly — Sedlák and Amis measured their static intensities exactly this way, on the same Brookhaven BI-200SM goniometer as their correlograms, "by counting the total number of counts per a defined time interval" [40]. The program will therefore build an SLS measurement from a set of DLS runs, one intensity per angle:

$$\bar{I}(\theta) = \frac{1}{N} \sum_{k=1}^{N} I_k(\theta) \tag{59}$$

where I_k are the samples of that run's count-rate trace. Where an instrument file records the run's totals instead of a trace, the same average follows from the header — for the Brookhaven `.dat`, the total counts divided by the number of correlator samples and the sampling time they were accumulated over [54]:

$$\bar{I} = \frac{N_{\text{counts}}}{N_{\text{samples}}\,\Delta t_{\text{sample}}} \tag{60}$$

The two agree to better than 1 % on the program's committed test data. Which one produced a given number is recorded on the measurement and travels into the export.

Two conditions the program cannot check for you, and so only states:

- **Comparability.** Intensities may be compared across runs only if the attenuation and the laser power were the same for all of them. An instrument that attenuates automatically between runs breaks this silently.
- **Detector linearity.** At high count rates the detector's dead time makes the measured rate fall below the true one. The correction is instrument-specific — for the BI-9000AT, $I_{\text{true}} = I_{\text{meas}}/(1 - \tau_{\text{d}} I_{\text{meas}})$ with $\tau_{\text{d}} = 25$ ns [55] — so consult your instrument's manual for its working range rather than assuming one. The program applies no such correction: the dead time depends on the detector, and it is not a number the program can know.

Where two or more runs share an angle — repeats at one setting — their mean count rates are averaged into a single point, through the same replicate-identity check the correlogram averaging uses (§6.4). Every contributing run is recorded, so the result does not depend on the order they were loaded in.

Uncertainty. A derived intensity carries no ±, and the trace mean's own sampling error is deliberately **not** propagated into the Zimm or Debye weighting. Two reasons. The sampling error of a count-rate average understates the real uncertainty in a static intensity, which is dominated by alignment, cell cleanliness and the calibration — so a ± built from it would read as precision the number does not have. And weighting a fit by it would change every result the platform already reports without an independent check that the weights are right. The honest route to a ± on these points is repeats, exactly as for any other static measurement. Propagating it instead would be a deliberate decision, and one that owes the Monte-Carlo check every propagated uncertainty in this program is validated against before it ships.

Because the derived measurement is an ordinary SLS measurement, everything downstream (§4.2–§4.5) applies to it unchanged, including the uncalibrated fallback above.

4.1.2 Mode-Resolved Intensities (the Sedlák Partition) When a sample relaxes by more than one mode, its static intensity is a sum of the modes' shares — and the DLS decomposition says how big each share is. Sedlák's method [43] splits the standard-normalized excess intensity by the modes' relative amplitudes, taken from the spectrum of relaxation times:

$$A_s(\theta) = \frac{I(\theta)/I_B(\theta)}{1 + A_f(\theta)/A_s(\theta)}, \qquad A_f(\theta) = \frac{I(\theta)/I_B(\theta)}{1 + A_s(\theta)/A_f(\theta)} \tag{61}$$

with the closure $I(\theta)/I_B(\theta) = A_f(\theta) + A_s(\theta)$. The same closure holds for any number of modes [44], and since the excess Rayleigh ratio is the normalized excess intensity times a fixed optical factor, the program partitions on the Rayleigh side, where the whole calibration pipeline (§4.1) is already behind it:

$$\frac{I(\theta)}{I_B(\theta)} = \sum_i A_i(\theta) \quad \Longrightarrow \quad \Delta R_i(\theta) = A_i(\theta)\,\Delta R(\theta) \tag{62}$$

Partitioning happens **after** solvent subtraction on purpose: the amplitudes decompose the polymer solution's correlated (excess) scattering, and solvent scatter belongs to no mode — splitting the raw intensity would allocate a share of the solvent baseline to each mode and then over-subtract it angle-dependently.

In the program, the DLS side owns the decomposition: a **mode set** is recorded from the Mode Map (§3.3) as a frozen snapshot — per-angle amplitude fractions with their source runs, extraction method, regularization and integration bounds — and the SLS tab's Mode-Resolved sub-tab partitions any intensity series of the sample (a static scan or a derived series, §4.1.1) by it. The fractions are used **as measured, never renormalized**: when the recorded modes do not exhaust the spectrum, the reported sum of fractions falls below 1 and the deficit is disclosed rather than silently reassigned. The method assumes the modes are *widely separated* on the time axis — Sedlák's own stated condition [43] — which the program checks as a ratio of adjacent decay rates against a documented house threshold (default 5); a strained separation is a note, never a refusal, and manually set integration bounds always carry the caveat (a manual split exists precisely because the peaks overlap).

The per-mode apparent size. Each mode's share is an ordinary angular series, and Sedlák's per-mode static analysis is a Guinier-type fit of the *slow-mode amplitude alone* [43] — the total-intensity form is valid only when one mode carries nearly all of the intensity, while the amplitude form survives any mixture because the fast mode's amplitude is angle-flat:

$$\frac{A_s(0)}{A_s(q)} \cong 1 + \frac{1}{3} R_{G,\mathrm{app}}^2\, q^2, \qquad q \to 0 \tag{63}$$

The program fits the exact rearrangement $1/A_s$ vs q^2, so the normalization $A_s(0)$ is the fit's own intercept rather than a hand-picked lowest angle, and $R_{G,\mathrm{app}}^2 = 3\,(\text{slope}/\text{intercept})$ — every intensity scale cancels, so the size survives an uncalibrated run and a missing $\mathrm{d}n/\mathrm{d}c$ alike. The subscript is Sedlák's own: the number is **apparent**, a semiquantitative estimate whose *relative changes* are the meaningful signal. Below a resolution floor computed from the fitted q-window (never copied from anyone's optics),

$$R_{G,\mathrm{floor}} = \sqrt{\frac{3\,\delta}{q_{\max}^2 - q_{\min}^2}} \tag{64}$$

with δ the documented minimum relative amplitude change (default 0.05), the predicted variation across the window is inside the scatter that ill-posed amplitudes carry, and the result is labeled relative-changes-only. The two-mode intensity-fraction decomposition itself ($A_f + A_s = 1$, per-mode contributions tracked against angle and temperature) is standard practice well beyond polyelectrolytes [56].

Any SLS analysis on a mode's share — flagged, never gated. A partitioned series is an ordinary Rayleigh series, so Guinier, Debye and (across concentrations) Zimm/Berry run on it unchanged, computed live in the Mode-Resolved sub-tab. Every such result is **apparent and mode-resolved**, and its Mw/A_2 carry three standing caveats the program states rather than

polices, because it cannot know which mode is the chains and which is solvent structure or domains: the $Kc/\Delta R$ ordinate uses the **total** concentration and the **chain** $\mathrm{d}n/\mathrm{d}c$, while a domain's own mass concentration and contrast are unknown — for a domain-type mode, scattering goes as the number, volume and contrast of the domains jointly [57], so an Mw computed with the chain increment is unanchored by an unverifiable contrast assumption, not just noisy. For charged systems, absolute Mw and A_2 additionally require dialysis-equilibrium (Donnan) conditions [44]. And near the theta point, the measured amplitude ratio mixes the gel and osmotic moduli [58] — a resolvable slow mode is itself the warning that $Kc/\Delta R$ is contaminated.

Uncertainty. Per-mode intensities carry no ± — each is one ill-posed inversion's amplitude times one intensity, the same stance as NNLS/CONTIN themselves (§6). The amplitude-Guinier $R_{G,app}$ reports the regression SE across independent angles (HC3, statistical only, *excluding* the inversion's own systematics — regularization and integration bounds), withheld whenever two fitted points share a source run, exactly as the per-mode dynamics fits treat the same amplitude class (§3.3). Because the size is the root of a coefficient ratio — a nonlinear transform — this SE carries its own Monte-Carlo validation in the amplitudes' own noise regime: it is **conservative**, meeting or slightly exceeding the sampling spread (measured ×1.15 at 3 % amplitude scatter, ×1.38 at 20 %), never under-reporting — the same safe direction as HC3 on the short A_2 ladder (§6.1). The ± on any per-mode Mw/A_2 excludes those same inversion systematics, and the result says so. The partition's provenance — mode set, method, α, bounds, source runs, date — travels into the affected export columns' Comments, so no per-mode number can be read without its origin.

4.2 Single-Concentration (Apparent) Analysis

At one concentration, a **Debye** plot [59] of $Kc/\Delta R_\theta$ against q^2 is linear; the intercept gives an apparent molecular weight and the slope an apparent radius of gyration. These are **apparent** because the intercept still contains the concentration term,

$$\left.\frac{Kc}{\Delta R_\theta}\right|_{q\to 0} = \frac{1}{M_w} + 2A_2 c, \tag{65}$$

i.e. $1/M_\mathrm{w} + 2A_2 c$, not $1/M_\mathrm{w}$. This single-concentration route still resolves angle, so it gives an apparent R_g; reducing further to a **single angle** leaves only an apparent M_w and no R_g. The result carries an *apparent* label to keep it distinct from the thermodynamic values of §4.3.

A point whose $Kc/\Delta R_\theta$ is **non-positive** — the solution scattering less than its solvent blank, the ordinary signature of an over-subtracted reference at high angle — has no place on either line: every linear fit here (Debye, Guinier, Zimm and Berry) drops such points before fitting and records how many, and a set left with fewer than two usable points is refused rather than fitted. The reliability of a size from any of these fits also carries its own flag: ${R_\mathrm{g}}^2$ is a ratio of two coefficients of the same regression, so a design too ill-conditioned for M_w is too ill-conditioned for R_g as well, and the value is shown with the flag rather than promoted silently into $\rho = R_\mathrm{g}/R_\mathrm{h}$.

4.3 Zimm and Berry Double Extrapolation

The full analysis combines several concentrations and angles and extrapolates to both zero angle and zero concentration. The first-order Zimm equation [1], [60],

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{M_w}\left(1 + \frac{q^2 R_g^2}{3}\right) + 2A_2 c, \tag{66}$$

is linear in (q^2, c), so a single global regression $Kc/\Delta R_\theta = a + b\ q^2 + d\ c$ gives all three at once:

$$M_w = \frac{1}{a}, \qquad R_g = \sqrt{\frac{3b}{a}}, \qquad A_2 = \frac{d}{2}. \tag{67}$$

The **Berry** construction [61], [62] uses the square-root ordinate $\sqrt{Kc/\Delta R_\theta} = a' + b'q^2 + d'c$, giving

$$M_w = \frac{1}{a'^2}, \qquad R_g = \sqrt{\frac{6b'}{a'}}, \qquad A_2 = d'\,a'. \tag{68}$$

Berry linearizes the high-qR_g regime (roughly M_w above ~1 MDa, or where qR_g approaches 1) better than Zimm. The program reports which construction it used, and the result is marked **thermodynamic**, not apparent.

> **Spacing constant.** The classic grid spreads each concentration's points along the x-axis by $q^2 + k\ c$, where k is a purely **aesthetic** spacing constant (User-Manual §10.2); the chosen k appears in the x-axis label. It only changes how spread-out the curves look — M_w, R_g, and A_2 come from the real (q^2, c) fit and are unaffected.

4.4 Calibration-Free Second Virial Coefficient

A_2 can be obtained without any absolute calibration, from ratios of excess intensities across concentration at a fixed angle. Because only ratios enter, the optical constant K and the Rayleigh-ratio calibration cancel:

$$Y(c) = \frac{I(c_\mathrm{ref})/c_\mathrm{ref}}{I(c)/c}, \qquad \frac{\text{slope}}{\text{intercept}} = 2A_2 M_w. \tag{69}$$

This yields the product $2A_2M_\mathrm{w}$ directly; if M_w is known **on an absolute scale**, A_2 follows. It is valuable when absolute calibration is uncertain, as in low-dn/dc systems.

That qualification is the whole point of the method, and it is easy to lose. The product is calibration-free because only intensity *ratios* enter, so the unknown scale cancels. Dividing by M_w puts the scale back in: an M_w from an uncalibrated run is not a molar mass but a number on an arbitrary scale, and A_2 computed from it is not an A_2. The program therefore reports A_2 only when the M_w behind it is trustworthy — from a calibrated run, or characterized independently and entered by hand — and **withholds it** otherwise, rather than printing a value that inherits the arbitrary scale. The product above it is reported either way.

A point whose excess Rayleigh ratio comes out **non-positive** — noise, or an over-subtracted solvent reference — cannot form $Y(c)$ and is dropped before the fit. The exported concentration and Y columns are the surviving points, so the loss would otherwise leave no trace in the file at all; the drop is therefore stated in the Comments cells of those columns and of the product. The Berry route (§4.3) drops non-positive $Kc/\Delta R$ the same way, before the square root, and says so in the same place.

Uncertainty. The SE on $2A_2M_w$ = slope/intercept is propagated from the HC3 covariance of the line (Eq. 92) through the ratio by the delta method (Eq. 97); the A_2 SE is the linear pass-through $/(2M_w)$, **with M_w treated as exact**. That last point matters more than it sounds: since an uncalibrated run's A_2 is obtained by supplying an independently characterized M_w by hand, the reported ± excludes that M_w's own uncertainty — which enters A_2 linearly, so a 5 % error on the supplied M_w is a 5 % error on A_2 that no ± here reports. Because this is a **ratio of two fitted coefficients** read off a **short concentration ladder**, it is the one platform SE where HC3's small-n conservatism is pronounced: on a typical 4–6-point ladder the reported ± errs on the cautious side and may exceed the true sampling spread, by roughly 20–40 %, tightening toward the exact value as more concentrations are added (≈ exact by 12–16 points). This is a property of the single-shot variance *estimate*, not of the propagation — the delta method itself is exact here (a generalized-least-squares fit given the true per-point variances recovers the sampling spread to < 1 %). The remedy for a tighter, unbiased ± is experimental: more concentration points, or repeat runs (§6.4). See §6.1.

4.5 Guinier Plot

A single-concentration route to R_g using the **log intensity** — the counterpart to the Debye plot of §4.2. The Guinier approximation $\Delta R(q) = \Delta R(0)\cdot\exp(-q^2 R_g^2/3)$ linearizes as

$$\ln[\Delta R(q)] = \ln[\Delta R(0)] - \frac{R_g^2}{3}\,q^2, \tag{70}$$

so a straight-line fit of $\ln(\Delta R)$ vs q^2 gives $R_g = \sqrt{-3\,\text{slope}}$ and an apparent M_w from the intercept ($\Delta R(0) = Kc\,M_w$). Like Debye's, the Guinier R_g is **scale-independent** (it comes from the slope of a log-intensity plot), so it is reliable even on an **uncalibrated** run; only the intercept-derived M_w needs calibration. The approximation is first-order — exact to fourth order in q for near-spherical particles [63] — so it is used strictly in the Guinier regime $qR_g \lesssim 1$ [60]; the program applies a slightly looser practical cutoff ($qR_g \lesssim 1.3$, overridable) and flags the largest qR_g used. A fit past a *measured* limit is a failed precondition of the method, so both the slope R_g and the intercept M_w lose their reliability flag and the note names the limit that was applied; the values are still reported. The opposite case is treated differently: when the angular dependence is too small to measure, the fitted slope is not negative, no R_g can be read, and qR_g is therefore unknown. That is the small-qR_g limit Eq. 70 assumes, so only R_g is withheld — the intercept M_w keeps its flag, and the note says which of the two cases occurred. A slope that is not merely flat but *rising* is a third case, and it is noted rather than flagged: no size is available there either, but the intercept is extrapolated against the trend, so the M_w read from it is likely low. Interparticle interference — a structure factor suppressed as $q \to 0$, as in concentrated or charged systems — and an over-corrected solvent blank both produce that shape, and the fit's notes name them. For larger species, Berry or Zimm linearize the high-qR_g regime better [1], [60].

4.6 Depolarization Correction (Anisotropic Scatterers)

An **optically anisotropic** scatterer — one whose polarizability differs along different molecular axes (rigid rods, many nanoparticles, small anisotropic molecules like toluene; flexible coils are very nearly isotropic) — scrambles the polarization of part of the light it scatters. With the **vertically polarized** incident light used by essentially every modern instrument, this appears as a horizontally polarized (*depolarized*) component. Measuring it both enables the **Cabannes**

correction that keeps M_{w} from being overestimated, and yields the **optical anisotropy** as a result in its own right. This is the **static** side of depolarized light scattering; the **dynamic** side (rotational diffusion D_{r} from the depolarized correlogram) is described in §3.4.

Convention. Two-letter subscripts are **(incident, analyzer)**: I_{VV} = vertical incident + vertical analyzer (the ordinary polarized intensity); I_{VH} = vertical incident + horizontal analyzer (the depolarized intensity). The **depolarization ratio** and its natural-light equivalent are

$$\rho_v = \frac{I_{\mathrm{VH}}}{I_{\mathrm{VV}}}, \qquad \rho_u = \frac{2\rho_v}{1+\rho_v}, \tag{71}$$

where ρ_{u} is the ratio for **unpolarized** incident light used by the classical literature (the conversion is exact). Physically $0 \le \rho_{\mathrm{v}} \le 3/4$: $\rho_{\mathrm{v}} = 0$ is an isotropic scatterer, $\rho_{\mathrm{v}} = 3/4$ the fully anisotropic limit. A value outside this range signals **stray depolarized light** (polarizer/analyzer leakage, reflections, or an over-subtracted dark count) and is flagged, not used.

Cabannes correction (vertical incidence). The anisotropic part inflates the measured VV Rayleigh ratio, so an uncorrected M_{w} comes out too high. The **isotropic** excess Rayleigh ratio — the one the Zimm/Debye/Berry analysis (§4.2–4.3) should use — is recovered by [64]

$$\Delta R_{\mathrm{iso}} = \Delta R_{\mathrm{VV}}\left(1 - \tfrac{4}{3}\,\rho_v\right). \tag{72}$$

For an isotropic scatterer ($\rho_{\mathrm{v}} = 0$) this is the identity, so applying it is always safe. The removed anisotropic fraction is $(4/3)\rho_{\mathrm{v}}$.

Optical anisotropy. In one standard parametrization [1] the apparent molecular weight and the depolarized Rayleigh ratio carry the mean-square optical anisotropy δ^2,

$$M_{\mathrm{app}} = M_w\left(1 + \tfrac{4}{5}\delta^2\right), \qquad \delta^2 = \frac{5\rho_v}{3-4\rho_v}, \tag{73}$$

so δ^2 is 0 for an isotropic scatterer and grows without bound as $\rho_{\mathrm{v}} \to 3/4$. Eliminating δ^2 between M_{app} and ρ_{v} reproduces Eq. 72 exactly — the two routes agree. (Anisotropy normalizations differ between conventions; convert through the depolarization ratio, which is convention-independent, rather than by equating differently-normalized anisotropies.)

Natural-light form (for older datasets). For **unpolarized** incident light the classical Cabannes factor [65] recovers the isotropic part from the total at 90°,

$$\Delta R_{\mathrm{iso}} = \Delta R_{\mathrm{total}} \cdot \frac{6-7\rho_u}{6+6\rho_u}. \tag{74}$$

This is provided for reading natural-light data; modern instruments use Eq. 72.

Uncertainty. A single VV/VH intensity pair carries **no** statistical SE on ρ_{v} (consistent with the project rule that one measurement is not an ensemble; §6). When **replicate** intensities are available, the SE on ρ_{v} is propagated as a ratio of independent quantities (Eq. 98); δ^2 and the Cabannes factor then follow by the delta method (Eq. 97).

4.7 Full-Angular Form-Factor Fit

The Zimm/Berry/Guinier expansions of §4.2–4.5 use only the **small-qR_g** limit of the particle form factor, $P(\mathrm{q}) \approx 1 - q^2 {R_g}^2/3$. The **form-factor fit** instead fits the *full* angular form factor to the ordinate [1], Eq. 2.1.8

$$\frac{Kc}{\Delta R(q,c)} = \frac{1}{M_w\, P(q;R_g)} + 2A_2 c, \tag{75}$$

so it stays valid to larger qR_g, where the linear-in-q^2 plots are biased by the form-factor curvature. Two models are offered [66]:

Solid sphere (returns the geometric radius R, with ${R_g}^2 = (3/5)R^2$, and also reports R_g) [63], [66]:

$$P(u) = \left[\frac{3(\sin u - u\cos u)}{u^3}\right]^2, \qquad u = qR. \tag{76}$$

Gaussian (Debye) coil (returns R_g) [59], [66]:

$$P(x) = \frac{2}{x^2}(e^{-x} - 1 + x), \qquad x = (qR_g)^2. \tag{77}$$

Both reduce to the Guinier limit $P \to 1 - q^2{R_g}^2/3$ as $qR_g \to 0$, so this fit **extends**, rather than replaces, the Guinier/Zimm slope. Because $P(\mathrm{q})$ is nonlinear in R_g, the parameters ($1/M_w$, R_g, and — across concentrations — A_2) are found by least squares, seeded from the linear Zimm/Debye estimate. A **single-concentration** fit is **apparent** (the $2A_2c$ term folds into the intercept, as in §4.2); a fit **across concentrations** is **thermodynamic** (A_2 resolved). As with the other SLS routes, R_g (and R) come from the angular shape and survive an **uncalibrated** run, while M_w/A_2 need calibration.

Sphere validity limit. The sphere $P(\mathrm{u})$ has its first minimum where its amplitude vanishes, $u = qR = 4.4934$ (the first root of $\tan u = u$, equivalently the first zero of the spherical Bessel j_1), i.e. $qR_g = 4.4934\sqrt{3/5} \approx 3.48$; this sphere intensity — first minimum near $qR \approx 4.5$, multi-lobe structure beyond — is plotted in [63, Fig. 6]. Past it $P(\mathrm{q})$ is multi-lobed and a single-sphere fit is ill-conditioned — the amplitude (M_w) and the zero location trade off, so M_w can be badly wrong even at $R^2 = 1$ (R^2 does not flag it). A sphere fit whose largest qR_g passes this minimum is therefore flagged and its M_w/A_2 marked unreliable (drop the highest angles, or use a full Mie treatment for high-contrast spheres); R_g is still reported. The multi-start least-squares (several qR_g-spread starts, lowest-cost kept) finds the global minimum within this range. The Debye coil $P(\mathrm{q})$ is monotonic, so it carries no such limit.

Size floor — the same limit from below. A fit is also flagged when its fitted size falls below **0.155 nm** as an R_g. The claim is only that the body is **smaller than any particle the measurement could have seen** — 0.155 nm as an R_g is a 0.4 nm sphere diameter, narrower than a water molecule. It is deliberately *not* "$P(\mathrm{q})$ is flat here, so M_w is meaningless": $P > 0.995$ across the whole angular range for any R_g below ≈ 3 nm, and there M_w is the ordinary, correct intercept, which the program rightly leaves reliable.

Where $P \to 1$ *does* matter is in explaining why neither bound above catches it: as the size goes to zero the shape term drops out of the ordinate, the largest qR_g is 0 and the Mie $P(q)$ is flat at 1, so a collapsed fit satisfies both. The floor closes that detection gap. It applies to every model, the monotonic coil included, and binds R_g for the sphere, coil, Mie and ellipsoid fits and the **outer radius** for the core-shell one (§4.10), whose contrast-weighted R_g is undefined whenever the contrasts cancel. The value is set so it means the same physical size as the DLS floor — §3.1's cumulant minimum R_h of 0.2 nm, a *radius*, which converts to an R_g of $0.2{\cdot}\sqrt{3/5} = 0.155$ nm; the DLS path *rejects* below it while this one flags.

Unlike the two bounds above, the floor also **withholds the size itself**. Their "R_g is still reported" holds because past a minimum it is the *amplitude* that trades off while the size stays sound; below the floor the size *is* the defect, so it is not offered as a R_g candidate for $\rho = R_g/R_h$ or the scaling plots. M_w, A_2 and the fitted numbers are all still shown, flagged.

Uncertainty. With distinct angles (genuinely independent points) the fit reports a statistical SE from the **HC3** covariance of the model Jacobian at the optimum (§6.1), propagated to $M_w = 1/(1/M_w)$ and $R = R_g\sqrt{5/3}$ by the delta method (Eq. 97) and labeled as **excluding** calibration/dn-dc systematics. The nonlinear $M_w = 1/(1/M_w)$ reciprocal is Monte-Carlo-validated against the sampling SD before it ships (§6); on a well-determined multi-angle × multi-concentration design the reported SE tracks the sampling spread to a few percent (mildly conservative).

4.8 Ellipsoid Form Factor and Shape Fitting

The sphere and coil of §4.7 are isotropic in size. An **ellipsoid of revolution** adds a **shape** parameter — the axial ratio ε (polar ÷ equatorial semi-axis) — fitted from the angular $P(q)$. For semi-axes $(R, R, \varepsilon R)$ the form factor is the orientation average of the sphere amplitude F (Eq. 76), computed numerically (Guinier; [66], Eq. 61):

$$P(q;R,\varepsilon) = \int_0^{\pi/2} F^2[q\,r(R,\varepsilon,\alpha)]\,\sin\alpha\,\mathrm{d}\alpha, \qquad r(R,\varepsilon,\alpha) = R\sqrt{\sin^2\alpha + \varepsilon^2\cos^2\alpha}. \tag{78}$$

The fit is parameterized by the radius of gyration (comparable to the other models), with the equatorial radius formed internally,

$$R_g^2 = \frac{R^2(2+\varepsilon^2)}{5}, \qquad R = R_g\sqrt{\frac{5}{2+\varepsilon^2}}. \tag{79}$$

At $\varepsilon = 1$ the integrand is constant and $\int_0^{\pi/2}\sin\alpha\,\mathrm{d}\alpha = 1$, so P collapses **exactly** onto the sphere (Eq. 76) with $R = R_g\sqrt{5/3}$; $\varepsilon > 1$ is **prolate**, $\varepsilon < 1$ **oblate**.

Degeneracy — the crux. A shape-from-scattering inverse is **not single-valued**: at low-to-moderate qR_g a prolate and an oblate ellipsoid (and a sphere) reproduce the *same* $P(q)$. The fit therefore returns a **list** of solutions — every axial ratio whose fit is comparable to the best (its extra RMS misfit below a small fraction of the data RMS) — and flags them **degenerate**, never silently choosing one. This mirrors the depolarized ellipsoid inverse (§3.5). As qR_g grows the prolate/oblate curves diverge and the fit resolves to one solution; a no-fit returns a single non-converged sentinel.

Selecting a degenerate solution (Cross-Sample). Because the scattering data cannot pick among a degenerate fit's solutions, each is offered in the Cross-Sample source pickers as an explicit,

fully-flagged choice (labeled by its distinguishing shape parameters), never an automatic default; picking one is guided manual entry for a user who has resolved the shape by an independent method, and the choice is flagged everywhere it propagates. When the solutions **agree on** R_g even though the shape is not resolved (the contrast-weighted size is determined while the radii are not — the common core–shell case), the picker additionally offers a single consensus value, the **mean** of the solutions' R_g. "Agree" is a documented tolerance: the peak spread $\max R_g - \min R_g$ is within **10 %** of the mean, **or** within **1 nm** absolute (the floor keeps the relative test from being needlessly strict at small sizes). This consensus value carries no statistical ± (a mean over degenerate solutions is not a regression estimate).

Axial-ratio identifiability. Because every shape shares the Guinier limit $P \to 1 - q^2 {R_g}^2/3$ as $qR_g \to 0$, the axial ratio is **unidentifiable** at small qR_g. The fit detects this from the sensitivity of the residual to ε (the ε column of the Jacobian) and, when ε is unresolved, **reports R_g but withholds the axial-ratio ±** and flags it — R_g and M_w are gated only on the core $(1/M_w,\ R_g)$ conditioning, so a well-determined size is not held hostage to an unresolvable shape.

Uncertainty. Each converged solution carries **its own** HC3 statistical SEs on R_g, the axial ratio ε, $M_w = 1/(1/M_w)$ and A_2 (§6.1), labeled as excluding calibration/dn-dc systematics; the **degeneracy is reported structurally** (the list of flagged solutions), never folded into a ±. The SEs are Monte-Carlo-validated against the sampling SD in a resolved regime before they ship (§6).

4.9 Full Mie Form Factor

The analytic sphere (Eq. 76) is the **Rayleigh-Debye-Gans** limit — valid only at low contrast ($|m - 1| \ll 1$) and small phase shift ($2x|m - 1| \ll 1$, §3.6). The **full Mie theory** ([53], §4.3–4.4) is the *general* homogeneous-sphere solution — any size parameter $x = 2\pi n_{med} R/\lambda_0$ and any relative index $m = n_{solute}/n_{med}$ — so it covers the classic aqueous-nanoparticle case (e.g. polystyrene latex in water, $m \approx 1.19$) that RDG and Rayleigh refuse. Here m enters the scattering **arithmetic**, not merely a validity gate, so the hand-entered **solute refractive index** is required (§3.6).

The Mie scattering coefficients are, in Riccati–Bessel form (permeability-matched; [53], Eqs. 4.56/4.57),

$$a_n = \frac{m\,\psi_n(mx)\,\psi_n'(x) - \psi_n(x)\,\psi_n'(mx)}{m\,\psi_n(mx)\,\xi_n'(x) - \xi_n(x)\,\psi_n'(mx)}, \qquad b_n = \frac{\psi_n(mx)\,\psi_n'(x) - m\,\psi_n(x)\,\psi_n'(mx)}{\psi_n(mx)\,\xi_n'(x) - m\,\xi_n(x)\,\psi_n'(mx)}, \tag{80}$$

with $\psi_n(z) = z\,j_n(z)$ and $\xi_n(z) = z\,h_n^{(1)}(z) = \psi_n(z) - i\,\chi_n(z)$ $(\chi_n(z) = -z\,y_n(z))$. The vertically-polarized amplitude function is ([53], Eq. 4.74)

$$S_1(\theta) = \sum_{n=1}^{n_{max}} \frac{2n+1}{n(n+1)} (a_n\,\pi_n + b_n\,\tau_n), \tag{81}$$

with the angle functions $\pi_n = P_n^1(\cos\theta)/\sin\theta$ and $\tau_n = \mathrm{d}P_n^1/\mathrm{d}\theta$ by their upward recurrences, and the series truncated at $n_{max} = \lceil x + 4x^{1/3} + 2 \rceil$ ([67]). Because a homogeneous sphere does **not** depolarize, the per-particle scattered intensity is $\propto |S_1|^2$ under either the **VV** or the **VU** (no-analyzer) geometry set in the Calibration panel — the two coincide for a non-depolarizing scatterer — and the **form factor** is the normalized angular intensity

$$P(q) = \frac{|S_1(\theta)|^2}{|S_1(0)|^2}, \qquad \cos\theta = 1 - \tfrac{1}{2}(q/k)^2, \quad k = \frac{2\pi n_{\text{med}}}{\lambda_0}, \quad x = kR, \tag{82}$$

with $R = R_g\sqrt{5/3}$ (a solid sphere) formed internally so the fit stays parameterized by R_{g}. This is fit to $Kc/\Delta R = 1/(M_{\text{w}}\ P(\text{q})) + 2A_2 c$ exactly as in §4.7.

Continuity — the ladder. As $m \to 1$, Eq. 82 collapses onto the sphere form factor (Eq. 76) with $u = qR$, and the per-particle intensity onto the RDG $R^6\ P(\text{qR})$ law of §3.6; as $x \to 0$ it collapses onto the Rayleigh R^6 law. So Mie composes as the top of the documented **Rayleigh → RDG → Mie** ladder, and where the contrast is within the RDG limit the program notes that the cheaper RDG model agrees. Mie extends the reliable range to high contrast (latex in water) that RDG and Rayleigh refuse, and its multi-start reaches the global fit basin even for large particles whose $P(\text{q})$ oscillates fast. But when the fitted $P(\text{q})$ dips to a **deep minimum** within the angular range (min P over the fitted angles < a documented floor, default 0.05), that angle's ordinate $1/(M_{\text{w}}\ P)$ is a noisy small difference of large intensities, so M_{w}/A_2 are ill-conditioned there (a wrong fit basin passes the condition-number and R^2 checks alone) — the fit **flags** them unreliable while still reporting the size R_{g} (constrained by the whole curve). This is the general-case analog of the sphere's past-first-minimum flag.

Uncertainty. The Mie $P(\text{q})$ fit carries HC3 statistical SEs on $M_{\text{w}} = 1/(1/M_{\text{w}})$, R_{g}, R and A_2 (§6.1), labeled as excluding calibration/dn-dc systematics and Monte-Carlo-validated against the sampling SD in both the thermodynamic and the apparent single-concentration regimes before they ship (§6). The relative index m is a fixed user input (treated as exact), so it contributes no ± term. As for the analytic models, R_{g} and R survive an uncalibrated run while M_{w}/A_2 need calibration.

The same Mie per-particle intensity $|S_1|^2$ also drives the DLS intensity → volume/number re-weighting in the Mie regime (§3.6), replacing the RDG refusal for high-contrast particles.

4.10 Core–Shell (Coated-Sphere) Form Factor

A **coated (core–shell) sphere** is a core of relative index $m_1 = n_{\text{core}}/n_{\text{med}}$ and radius $R_1 = R_{\text{core}}$ inside a concentric shell of index $m_2 = n_{\text{shell}}/n_{\text{med}}$ out to $R_2 = R_{\text{outer}}$ (size parameters $x = kR_1$, $y = kR_2$; $k = 2\pi n_{\text{med}}/\lambda_0$). Both refractive indices are hand-entered — the shell is a second solute refractive index on the Data tab — and neither is ever guessed. The scattered field has the **same** form as the homogeneous sphere (§4.9): only the scattering coefficients change, from the Aden–Kerker generalization of the Mie coefficients ([68]; [53], §8.1 Eq. 8.2),

$$a_n = \frac{\psi_n(y)\,[\psi_n'(m_2y) - A_n\chi_n'(m_2y)] - m_2\,\psi_n'(y)\,[\psi_n(m_2y) - A_n\chi_n(m_2y)]}{\xi_n(y)\,[\psi_n'(m_2y) - A_n\chi_n'(m_2y)] - m_2\,\xi_n'(y)\,[\psi_n(m_2y) - A_n\chi_n(m_2y)]}, \tag{83}$$

$$b_n = \frac{m_2\,\psi_n(y)\,[\psi_n'(m_2y) - B_n\chi_n'(m_2y)] - \psi_n'(y)\,[\psi_n(m_2y) - B_n\chi_n(m_2y)]}{m_2\,\xi_n(y)\,[\psi_n'(m_2y) - B_n\chi_n'(m_2y)] - \xi_n'(y)\,[\psi_n(m_2y) - B_n\chi_n(m_2y)]}, \tag{84}$$

where the ratio functions A_n, B_n carry the **core** contribution and depend on the core size x only,

$$\begin{aligned} A_n &= \frac{m_2\,\psi_n(m_2x)\,\psi_n'(m_1x) - m_1\,\psi_n'(m_2x)\,\psi_n(m_1x)}{m_2\,\chi_n(m_2x)\,\psi_n'(m_1x) - m_1\,\chi_n'(m_2x)\,\psi_n(m_1x)}, \\ B_n &= \frac{m_2\,\psi_n(m_1x)\,\psi_n'(m_2x) - m_1\,\psi_n(m_2x)\,\psi_n'(m_1x)}{m_2\,\chi_n'(m_2x)\,\psi_n(m_1x) - m_1\,\psi_n'(m_1x)\,\chi_n(m_2x)}. \end{aligned} \tag{85}$$

These are evaluated by the numerically stable **BHCOAT** recurrence ([53], Appendix B): the logarithmic derivatives $D_n = \psi_n'/\psi_n$ by the downward recurrence of §4.9 (pole-free), the Riccati–Bessel ψ_n, χ_n upward, the series truncated at $n_{\max} = \lceil y + 4y^{1/3} + 2 \rceil$ on the **outer** y, and an inner-sphere convergence test that freezes the coefficients onto the homogeneous outer sphere once the core contribution is negligible. The amplitude function $S_1(\theta)$ (Eq. 81) and the form factor are then exactly as in §4.9:

$$P(q) = \frac{|S_1(\theta)|^2}{|S_1(0)|^2}, \qquad \cos\theta = 1 - \tfrac{1}{2}(q/k)^2, \quad x = kR_1, \quad y = kR_2. \tag{86}$$

Continuity. When $m_2 \to m_1$ (or $R_1 \to 0$), $A_n = B_n = 0$ and Eq. 86 collapses onto the homogeneous Mie form factor (Eq. 82) of the outer sphere — so core–shell composes on top of the Rayleigh → RDG → Mie ladder.

Reported size — the contrast-weighted radius of gyration. A two-density object's small-q (Guinier) radius of gyration is the contrast-weighted second moment [69] (core contrast $\propto m_1 - 1$, shell contrast $\propto m_2 - 1$; the common n_{med} factor cancels):

$$R_g^2 = \frac{3}{5}\,\frac{(m_1-1)\,R_1^5 + (m_2-1)\,(R_2^5 - R_1^5)}{(m_1-1)\,R_1^3 + (m_2-1)\,(R_2^3 - R_1^3)}. \tag{87}$$

This is the R_g the coated $P(q)$ exhibits in its Guinier limit — directly comparable to the Zimm/sphere R_g. It is **undefined** (reported blank) for a **contrast-matched** object, where the denominator (the net excess scattering length) is near zero or $R_g^2 \le 0$; there a single scattering size is not physically defined.

Degeneracy and uncertainty. Two radii from one angular curve are frequently **underdetermined**: at low qR_g every geometry shares the Guinier limit, and several (R_1, R_2) pairs reproduce the same $P(q)$. The fit therefore returns a **list** of comparably-fitting solutions, each flagged `degenerate` (like the ellipsoid, §4.8), and the degeneracy lifts only at high qR_g where the coated oscillations separate core from outer. It carries HC3 statistical SEs on $M_w = 1/(1/M_w)$, the two fitted radii ($R_2 = R_1 + t$ a linear combination) and A_2 (§6.1) — Monte-Carlo-validated against the sampling SD in both the thermodynamic and apparent regimes (§6) — while the contrast-weighted R_g (Eq. 87) carries **no** ±: it is a nonlinear combination of a structurally-degenerate radius pair, so its uncertainty is reported by the solution list, not a symmetric interval. The contrasts m_1, m_2 are fixed user inputs (exact, no ± term); the radii and R_g survive an uncalibrated run while M_w/A_2 need calibration. A fit whose $P(q)$ spans a **deep form-factor minimum** flags M_w/A_2 unreliable exactly as the Mie sphere does, and so does one whose **outer radius** falls below the size floor (§4.7) — the one place that floor binds a radius rather than R_g, because this model's R_g is undefined whenever the contrasts cancel.

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5. Solvent Property Library

Every scattering calculation needs the solvent's **refractive index** $n(\lambda, T)$ — it sets the scattering vector q (Eq. 1) — and its **dynamic viscosity** $\eta(T)$ — it sets the Stokes–Einstein radius R_h (Eq. 2). The library **proposes** both from a chosen solvent, temperature, and wavelength, as an

overridable convenience default: the value is provenance-tagged and reaches the analysis only when you commit it.

Unified forms. One functional form per property serves every solvent. The refractive index is a three-term Cauchy dispersion [70] with a linear temperature correction to a reference temperature T_{ref} (λ in µm) [71]:

$$n(\lambda, T) \;=\; A \;+\; \frac{B}{\lambda^2} \;+\; \frac{C}{\lambda^4} \;+\; \frac{\mathrm{d}n}{\mathrm{d}T}\,(T - T_{\mathrm{ref}}) \tag{88}$$

The viscosity is the Vogel–Fulcher–Tammann (VFT) form (η in Pa·s) [72]:

$$\eta(T) \;=\; A_\eta \exp\!\left(\frac{B_\eta}{T - T_0}\right) \tag{89}$$

The coefficients are fitted to each property's authoritative correlation over its validity range; outside that stored wavelength/temperature range the program does not guess — it declines the value rather than extrapolating. Each solvent carries a confidence tier — `primary` (dispersion and reference-correlation data) or `estimate` (weaker, e.g. bulk-compilation data) — shown as a display badge. The full per-solvent source list is in the Citation Index (Doc 5).

Uncertainty: shown, not propagated. Each property carries a display-only tolerance — an absolute n uncertainty, a relative η uncertainty — evaluated at the requested wavelength and temperature rather than as one flat figure. It informs your judgement but is **not** folded into any fitted standard error: the reported ± is statistical/regression uncertainty, which excludes calibration-class systematics (§6.5), and a proposed n/η is exactly that class. The refractive-index tolerance combines the source accuracy, the unified-form fit residual, the $\mathrm{d}n/\mathrm{d}T$ slope uncertainty, and an inter-laboratory allowance in quadrature:

$$\sigma_n(\lambda, T) \;=\; \sqrt{u_{\mathrm{src}}^2 \;+\; r_\lambda(\lambda)^2 \;+\; r_T(T)^2 \;+\; \left(u_{\mathrm{slope}}\,(T - T_{\mathrm{ref}})\right)^2 \;+\; u_{\mathrm{lab}}^2} \tag{90}$$

and the viscosity tolerance likewise — its $u_{\mathrm{lab},\eta}$ term applies only where the source is not a critically evaluated reference correlation (an evaluated correlation has already done the inter-laboratory analysis):

$$\sigma_{\eta,\mathrm{rel}}(T) \;=\; \sqrt{u_{\mathrm{src,rel}}^2 \;+\; r_{T,\eta}(T)^2 \;+\; u_{\mathrm{lab},\eta}^2} \tag{91}$$

The u_{lab} terms are fixed allowances for how much independent laboratories' published measurements of the same property typically disagree — the systematic a single source's stated accuracy cannot contain. They are measured, not chosen: $u_{\mathrm{lab}} = 1.2 \times 10^{-3}$ is the median, over wavelength nodes, of the full spread between the published $n(\lambda,\ T)$ curves covering that node, and $u_{\mathrm{lab},\eta} = 3.2\,\%$ is half the median 10th-to-90th-percentile spread of published viscosity measurements (≈ 1.28 standard deviations, adopted as-is). The band remains display-only. It is flat in λ apart from the fit-residual envelope, so at the blue edge of a record's wavelength range — where dispersion sources are typically weakest — it can still be somewhat optimistic.

The one notable case is water, whose $\mathrm{d}n/\mathrm{d}T$ is genuinely temperature-dependent (the near-20 °C slope steepens by roughly two-thirds by 45 °C). Its stored coefficients are fitted directly to the

IAPWS refractive-index formulation [73] evaluated densely across 400–800 nm and 15–60 °C, with the 0.1 MPa liquid density from the IAPWS-95 equation of state [74], so the drift is carried per condition by the residual envelope r_T rather than by an inflated slope allowance. One disclosure: [73] states its accuracy as 1.5×10^{-5} in the visible but only 1×10^{-3} between 700 and 800 nm, so the red edge of the water band rests on weaker truth there — the inter-laboratory allowance (1.2×10^{-3}) covers it.

Solvent caveats. Some error sources sit outside what the program can measure. The clearest is **glycerol**, which is strongly hygroscopic: its viscosity rises steeply as adsorbed water is excluded, so the tabulated η is for dry glycerol and a real, water-equilibrated sample can differ by tens of percent [72]; its refractive index shifts with water uptake as well [71]. **DMSO and DMF** are hygroscopic too, and their viscosity sources measured dried, water-quantified samples — a bench-equilibrated sample can sit outside the tabulated band, in a direction and by an amount the band cannot know.

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6. Uncertainty Estimates

The program reports the **statistical** (regression) uncertainty of a fit, where the fitted points are genuinely independent measurements. Where no honest uncertainty exists, none is reported. Two scope rules follow from this:

1. **Statistical only.** Every ± below is the spread implied by the scatter of the fitted points. It **excludes systematic error** — calibration (the constant k_c) and $\mathrm{d}n/\mathrm{d}c$ in particular. Because $M_w \propto 1/(\mathrm{d}n/\mathrm{d}c)^2$ and scales with k_c, those systematics usually dominate the true error of an absolute M_w or A_2; the displayed ± is a **lower bound** on the total uncertainty and is labeled *"± statistical; excludes calibration & $\mathrm{d}n/\mathrm{d}c$"* wherever a calibration-dependent quantity is shown. R_g, R_h, the size exponent ν, and the calibration-free $2A_2M_w$ are scale-independent, so their statistical ± is the whole statistical story. When $\mathrm{d}n/\mathrm{d}c$ is **absent** rather than merely uncertain (§1.4), the same split holds in its strongest form: the scale-free quantities keep a fully meaningful ±, while the ± on M_w and A_2 describes only the scatter about an arbitrary scale and travels beside a flag saying so. No uncertainty is invented for the missing factor — an unknown multiplicative constant has no distribution the data can speak to.
2. **No uncertainty from a single correlogram.** A correlogram's lag channels are strongly correlated [29], so an ordinary least-squares covariance from **one** correlogram assumes independence it does not have and **under-reports** the truth. ISO 22412 [27] therefore derives DLS dynamic uncertainty from **repeat measurements**. Accordingly the single-correlogram dynamic fits (cumulant, single/double exponential, KWW, lognormal) report **no ±**. The honest dynamic ± instead comes from **replicate averaging** (§6.4): when you average true repeats, the spread of the per-replicate fits gives an ISO 22412 SD/SEM. The cumulant PDI and the fit RMS remain as width/goodness diagnostics.

6.1 The estimator: HC3 heteroscedasticity-consistent standard errors

For a linear model $\mathbf{y} = \mathbf{X}\mathbf{b}$ with n points, p parameters and residuals e_i, the parameter covariance is the **HC3 "sandwich"** estimator — the simplified form of Long & Ervin [75] (MacKinnon & White [76] coined "HC3" as the jackknife variant with an extra $(n-1)/n$ factor, which we do not use):

$$\mathrm{Cov}(\mathbf{b}) = (\mathbf{X}^\top\mathbf{X})^{-1}\left[\sum_i \frac{e_i^2}{(1-h_{ii})^2}\,\mathbf{x}_i\mathbf{x}_i^\top\right](\mathbf{X}^\top\mathbf{X})^{-1}, \qquad h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i. \tag{92}$$

Unlike the classical $s^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$, HC3 does **not** assume the points share one error variance, so it holds up when point precision varies across angles or concentrations (e.g. a per-angle Γ whose precision changes with q). It reduces to the ordinary formula under equal-variance errors and is mildly **conservative** at small n (Monte-Carlo-validated; §6.5). When the residual degrees of freedom are too few ($n - p < 2$) no SE is reported, since one residual cannot honestly support a spread (and HC3 blows up as $n - p \to 1$); p counts every parameter the fit consumed, including one later profiled out (the ellipsoid form factor's unresolved axial ratio), under both estimators. The standard error of a coefficient is the square root of its diagonal, e.g. for a straight line $y = a + b\,x$:

$$\mathrm{SE}(b) = \sqrt{\mathrm{Cov}_{bb}}, \qquad \mathrm{SE}(a) = \sqrt{\mathrm{Cov}_{aa}}. \tag{93}$$

A variance of exactly zero is **withheld**, not reported as ± 0 — a standard error of zero claims infinite precision, which no measurement supports. This is the same rule §6.5 states for a variance that cancels to zero or below, applied to each diagonal entry as it is taken, and the same rule §6.4 applies to a zero spread across repeats.

Two things reach a zero variance, and the program does not try to tell them apart. A fit through **exactly** collinear data — a synthetic set built from the model it is fitted with — can leave residuals that are zero to the last bit rather than at the ~10^{-15} rounding floor. Separately, a variance can **underflow** to zero when the abscissa is extreme enough that the covariance arithmetic falls off the bottom of the floating-point range. The first is honest but carries no information; the second would be an under-report. Both leave the ± blank, and in the exported file that is an empty cell — the program's convention for a missing value throughout.

The conservatism of the HC3 SE at small n is largest for a **ratio of two fitted coefficients on a short, high-leverage design** — the calibration-free $2A_2M_\mathrm{w}$ = slope/intercept (§4.4) is the platform's clearest case: at a 5-point ladder the endpoints carry leverage ≈ 0.5–0.7, HC3 up-weights their residuals by $(1-h)^{-2}$, and the reported ± runs ~20–40 % above the true sampling spread. This is *only* the residual-based variance estimate being cautious — the underlying delta-method propagation is exact (a GLS fit given the true per-point variances matches the sampling spread to < 1 %), and it is not a shared-reference/common-mode effect (fitting the equivalent unshared form $c/\Delta R(c)$ gives an identical SE). The ± is therefore honest-but-loose on the cautious side; it tightens toward exact as the concentration ladder lengthens, and an unbiased ± comes from replicates (§6.4). Where a single-shot SE rests on few, high-leverage points, the achievable honest target is *conservative and disclosed*, with more points or replicates the route to an unbiased one.

A through-origin slope $y = b\,x$ (used for $\Gamma = D\,q^2$) uses the same sandwich with the one-column design $\mathbf{X} = \mathbf{x}$, giving $\mathrm{SE}(b)$ from the scalar covariance.

The Zimm/Berry global fit is multilinear — ordinate = $a + b\,q^2 + d\,c$ with design $\mathbf{X} = [1,\ q^2,\ c]$ — and uses the same $\mathrm{Cov}(\mathbf{b})$ of Eq. 92 for the full 3×3 covariance:

$$\text{ordinate} = a + b\,q^2 + d\,c, \qquad \mathbf{X} = [\,\mathbf{1}\ \ \mathbf{q^2}\ \ \mathbf{c}\,]. \tag{94}$$

How the inverse is computed. Every equation above is written with $(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$, but $\mathbf{X}^\mathsf{T}\mathbf{X}$ is never formed. Forming the normal equations explicitly loses information in finite-precision arithmetic [35]: the singular values of $\mathbf{X}^\mathsf{T}\mathbf{X}$ are the *squares* of those of $\mathbf{X}$, so the inverse costs two significant digits per decade of condition number instead of one. Scaling each column of $\mathbf{X}$ to unit norm — giving $X = \hat{X}D$ with $\mathbf{D}$ the diagonal of column norms — and taking the SVD $\hat{X} = USV^\top$ gives every quantity the estimators need, exactly:

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \mathbf{D}^{-1}\mathbf{V}\mathbf{S}^{-2}\mathbf{V}^\top\mathbf{D}^{-1}, \qquad h_{ii} = \|\mathbf{U}_i\|^2. \tag{95}$$

The leverages come from the same decomposition because h_{ii} is a diagonal entry of the orthogonal projector onto the column space of $\mathbf{X}$, which column scaling leaves unchanged. Taken this way they satisfy $0 \le h_{\mathrm{ii}} \le 1$ and $\Sigma h_{\mathrm{ii}} = p$ by construction, and each covariance is assembled as a product $\mathbf{B}\mathbf{B}^\mathsf{T}$, so it is positive semi-definite whatever the rounding — a variance can never come back negative. This is a numerical choice only: the estimator is Eq. 92 either way, and on a well-conditioned design the standard errors agree with the direct computation to about 4 parts in 10^{15} — some twenty times the last bit, and far below any uncertainty the number itself carries.

6.1.1 Classical OLS standard errors (opt-in, for comparability) Settings offers a switch from HC3 (the default) to the **classical homoscedastic OLS** covariance [77], the one reported by textbooks, classical light-scattering software and spreadsheets:

$$\mathrm{Cov}_{\mathrm{OLS}}(\mathbf{b}) = s^2\,(\mathbf{X}^\top\mathbf{X})^{-1}, \qquad s^2 = \frac{\sum_i e_i^2}{n-p}. \tag{96}$$

This assumes every point shares one error variance. It is intended for reproducing a result from classical software or a published table *like-for-like*, rather than as a general improvement. On a short, high-leverage design it **under-reports**: on the 5-point calibration-free A_2 ladder above, OLS returns ≈ 0.90 of the true sampling spread, where HC3 sits at ≈ 1.3 on the cautious side; the OLS deficit shrinks toward 1.0 as points are added. Under-reporting is the direction the platform avoids by default, which is why **HC3 is the default and OLS is a labeled, recorded opt-in**. The selected estimator is recorded on every result that carries a regression ± and written into the export — an OLS ± column's *Comments* cell reads `SE: classical OLS` (HC3 is the silent default). The switch changes only the ±; point estimates (M_{w}, R_{g}, D, R_{h}) are identical either way. Everything downstream (the delta method of §6.2, the ratio and power-law SEs) propagates whichever covariance was selected.

6.2 Propagating a covariance to a derived quantity (delta method)

A scalar function $f(\mathbf{b})$ of the fitted coefficients has, to first order, the standard error given by the **law of propagation of uncertainty** [78]:

$$\mathrm{SE}(f) = \sqrt{\mathbf{J}^\top\,\mathrm{Cov}(\mathbf{b})\,\mathbf{J}}, \qquad \mathbf{J} = \nabla_{\mathbf{b}} f. \tag{97}$$

This delivers every SLS SE from Eqs. 92, 94 with the appropriate Jacobian $\mathbf{J}$:

Quantity	Relation	Jacobian **J** (order of params)
M_{w} (Zimm)	$M_{\mathrm{w}} = 1/a$	$[-1/a^2,\ 0,\ 0]$
A_2 (Zimm)	$A_2 = d/2$	$[0,\ 0,\ 1/2]$
R_{g} (Zimm)	$R_{\mathrm{g}} = \sqrt{3b/a}$	$[-R_{\mathrm{g}}/(2a),\ R_{\mathrm{g}}/(2b),\ 0]$
M_{w} (Berry)	$M_w = 1/a'^2$	$[-2/a'^3,\ 0,\ 0]$
A_2 (Berry)	$A_2 = d'a'$	$[d',\ 0,\ a']$
$M_{\mathrm{w}}^{\mathrm{app}}$ (Debye)	$= 1/\text{intercept}$	$[-1/\text{intercept}^2,\ 0]$
$R_{\mathrm{g}}^{\mathrm{app}}$ (Debye)	$= \sqrt{3\,\text{slope}/\text{intercept}\cdot M_w}$	$[-R_{\mathrm{g}}/(2{\cdot}\text{intercept}),$ $R_{\mathrm{g}}/(2{\cdot}\text{slope})]$
R_{g} (Guinier)	from slope of $\ln \Delta R$ vs q^2	$\mathrm{SE}(R_{\mathrm{g}}) = (3/2R_{\mathrm{g}}){\cdot}\mathrm{SE}(\text{slope})$
$2A_2M_{\mathrm{w}}$ (cal-free)	$= \text{slope}/\text{intercept}$	$[-\text{slope}/\text{intercept}^2,$ $1/\text{intercept}]$

For a **ratio** of two *independent* quantities $r = a/b$ (used for $\rho = R_{\mathrm{g}}/R_{\mathrm{h}}$, with R_{g} from SLS and R_{h} from DLS — measured independently), Eq. 97 reduces to the familiar fractional sum in quadrature:

$$\mathrm{SE}(r) = \left|\frac{a}{b}\right| \sqrt{\left(\frac{\mathrm{SE}(a)}{a}\right)^2 + \left(\frac{\mathrm{SE}(b)}{b}\right)^2}. \tag{98}$$

ρ is given a $\pm$ **only when both** R_{g} and R_{h} carry an SE (i.e. both came from a regression — a Zimm/Berry R_{g} with a Γ-q^2 or D-c R_{h}); a single-cumulant R_{h} has no SE, so ρ then prints without $\pm$.

For a **power map** $f = k\,x^{\mathrm{p}}$ (used for $R_{\mathrm{h}} \propto D^{-1}$, so $p = -1$), Eq. 97 gives a fractional relation:

$$\frac{\mathrm{SE}(f)}{|f|} = |p|\,\frac{\mathrm{SE}(x)}{|x|}, \qquad \text{so } \frac{\mathrm{SE}(R_h)}{R_h} = \frac{\mathrm{SE}(D)}{D}. \tag{99}$$

Both the ratio (Eq. 98) and the reciprocal power map (Eq. 99) are **first-order** results, exact only while the *denominator's* fractional error is small. Monte-Carlo checks against the sampling spread confirm they track it to a few percent at realistic precision (a denominator coefficient of variation $\lesssim 0.05$, where a multi-angle Γ-q^2 fit sits) and turn mildly **anti-conservative** — under-reporting by roughly 5–10 % — once the denominator CV reaches ≈ 0.15, with the heavy tail of $1/(\text{denominator})$ taking over beyond ≈ 0.3. The direction is set by which quantity is noisy: $\rho = R_{\mathrm{g}}/R_{\mathrm{h}}$ at a viscous-solvent R_{h} CV of 0.10–0.15 under-reports a few percent, whereas $\rho_{\mathrm{v}} = I_{\mathrm{VH}}/I_{\mathrm{VV}}$ is immune (its noisy channel is the numerator). As everywhere in §6, the honest remedy for a weak denominator is **more points or replicates** — a smaller denominator CV — not a wider algebraic formula. The reciprocal R_{h} SE (Eq. 99) is carried unchanged wherever R_{h} follows from D: the Γ-q^2 and D-c extrapolations, and the depolarized $R_{\mathrm{h,t}}$ from D_{t}.

The cross-sample scaling exponents (ν in $R_{\mathrm{g}}/R_{\mathrm{h}} \propto M_{\mathrm{w}}{}^{\nu}$, the slope in $A_2 \propto M_{\mathrm{w}}{}^{\text{slope}}$) are slopes of a log–log regression, so their SE is Eq. 93 applied to ($\log M_{\mathrm{w}}$, $\log y$).

6.3 A consistency check, not an uncertainty: the Zimm two-route M_{w}

The Zimm grid is extrapolated to the intercept along two independent routes — the $c \to 0$ line and the $q \to 0$ line — which must meet at the same $1/M_{\mathrm{w}}$. The program reports both intercept-derived

weights and their relative gap:

$$\text{agreement} = \frac{\left|M_w^{(c\to 0)} - M_w^{(q\to 0)}\right|}{\frac{1}{2}\left(M_w^{(c\to 0)} + M_w^{(q\to 0)}\right)}. \tag{100}$$

This is a **diagnostic of extrapolation quality** (curvature, an over-masked grid), not a statistical uncertainty; a gap above ~10 % is flagged for inspection (the interface phrases it as the two routes "differ by X %"). On an **uncalibrated** run the two weights themselves are withheld — they carry the arbitrary scale the panel has already declined to print — while the gap, being a ratio, is unaffected and still reported, so the check works either way. On the Zimm/Berry plot both extrapolation lines are drawn **all the way to the axis** ($q^2 \to 0$ and $c \to 0$), meeting at the common intercept ($\approx 1/M_\mathrm{w}$, marked), so the double extrapolation is visible end to end.

6.4 Repeat measurements: a DLS dynamic uncertainty from replicates (ISO 22412)

Because a single correlogram cannot supply an honest dynamic ± (the single-correlogram scope rule above), the program gets one the way ISO 22412 [27] prescribes — from **repeat measurements**. Select a set of true replicates (the same sample re-measured back-to-back, e.g. a 10-run repeat set) and run **Average derived results** (User-Manual §10.1).

Both averaging actions check that the selection really is a repeat set, and refuse it with the offending field named when it is not — as does the DDLS analysis, which averages same-angle same-polarization correlograms of its own accord (§3.4). The runs must agree on solute and solvent, temperature (to 0.1 °C, the same bucket the sidebar groups samples by), scattering angle, wavelength, concentration, M_w fraction and analyzer geometry. Viscosity and the refractive indices are not compared: the program cannot tell whether a solvent property matches the solvent it is labeled with, so a mismatched one gives a wrong size whether or not the runs are averaged, and those parameters are yours to set and to check. Note that an averaged measurement inherits all three from the first run in the selection. Concentration and angle are the two worth knowing about, because a concentration series is deliberately one sample and an angular scan is deliberately one measurement set: the decay rate depends on concentration, so a mean across a ladder describes no real sample, and angular dependence has its own analysis (§3.2) rather than being a spread to average away. Average each condition separately.

Each replicate is fitted independently with the method you pick, and for a parameter x (e.g. R_h) the n fitted values are summarized as a mean and a standard error of the mean [78, Sec. 4.2]:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \mathrm{SEM} = \frac{s}{\sqrt{n}}. \tag{101}$$

The reported value is $\bar{x} \pm \mathrm{SEM}$ (none for $n < 2$, where no spread is defined). If the replicates agree exactly, $s = 0$, and the ± is **withheld** rather than reported as ± 0 — the same rule §6.1 states for a zero fit variance, and for the same reason: a standard error of zero claims infinite precision. The agreement is real, but it is a claim about the arithmetic rather than about the sample, and in practice it means the set is not made of independent repeats — a zero-noise synthetic set, or the same run loaded twice. The mean and n are still reported. Unlike a within-correlogram covariance, the x_i from separate runs **are** independent draws, so Eq. 101 does not under-report; it also captures genuine run-to-run drift (alignment, dust, number fluctuations) that one curve cannot see.

The averaged $R_h \pm$ SEM reaches the sample under one condition and is withheld under several:

- **Written** into the sample (so $\rho = R_g/R_h$ can use it) when at least two replicates fitted **completely** — every parameter of the chosen method finite, so one bad parameter drops the whole run. It never overwrites a hand-entered R_h (an explicitly **picked** R_h is replaced — pressing Run is a deliberate act — and the summary reports the replacement).
- **Not written — and the summary says which** — when there are too few complete fits, the method resolves more than one R_h mode, the sample already carries a hand-entered value, the averaged R_h is not a finite number, or one or more of the averaged fits returns a size outside the range DLS can establish (§3.1). That last case declines the write rather than qualifying it: the mean and its spread are still reported, but a size beyond what the technique resolves must not reach $\rho = R_g/R_h$ carrying a $\pm$, which would read as precision. Set the sample's R_h yourself if you mean to use it.

This SE was Monte-Carlo validated: across many synthetic n-replicate sets the reported SD/$\sqrt{n}$ matched the actual SD of the set-means to ≈ 1 % and the mean estimate was unbiased.

The companion action, **Average correlation functions**, instead takes the channel-by-channel mean of the replicate $g_2(\tau) - 1$ curves (requiring an identical lag grid) to produce a cleaner *point-estimate* correlogram you can fit like any measurement; it carries no $\pm$ of its own — the $\pm$ comes from Eq. 101.

Multi-peak (distribution) methods. *Average derived results* also accepts the distribution methods (NNLS / CONTIN / lognormal), which resolve a *variable* number of size-distribution peaks. These are averaged **by order of appearance**: each replicate's peaks are sorted by R_h, and position 1 (smallest R_h) is averaged across all runs, position 2 across all, etc., each reported as mean $\pm$ SD/$\sqrt{n}$ (Eq. 101) with its mean intensity weight and how many of the N runs resolved it. Two cautions are built in: (i) when runs **disagree on the peak count** a warning is shown, because a run that misses a peak shifts the positional alignment (inspect the distribution overlay before trusting the per-peak averages); and (ii) NNLS/CONTIN are **ill-posed** — their peak *positions* depend on the regularization and their fine structure is not always reproducible — so the cross-run $\pm$ is an honest **reproducibility spread**, not a calibrated error, and these peaks are **report-only** (never written to the sample's R_h). To get an averaged multi-population R_h that *can* feed ρ, average the correlograms (above) and fit that single clean curve with CONTIN in the usual way.

6.5 Where uncertainties are and are not reported

Reported (independent points → regression SE valid): SLS Zimm/Berry M_w, R_g, A_2; Debye/Guinier apparent M_w, R_g; calibration-free $2A_2M_w$ (and A_2 if an absolute-scale M_w is supplied — §4.4); Γ-vs-q^2 D and R_h; D-vs-c D_0, $R_{h,0}$, k_D; the scaling exponents; $\rho = R_g/R_h$ when both inputs have an SE; and the **replicate-averaged DLS dynamic** parameters (Eq. 101 — the DLS dynamic $\pm$ obtainable at a single angle and concentration, where no regression is available).

Not reported (no honest statistical uncertainty): every single-correlogram dynamic fit (the single-correlogram rule above), *unless* repeats are averaged via §6.4; the NNLS/CONTIN distributions (ill-posed inversions — no defensible covariance; the "fine structure is not always reproducible" caveat stands); a single-angle M_w (one datum, no fit); and a fit whose design is **too near-collinear to carry a covariance** — specifically, one whose condition number κ (§6.7) exceeds 10^{12}, at which point fewer than about four significant digits of the covariance survive. No $\pm$ is then shown for any parameter of that fit, an exactly rank-deficient design ($\kappa = \infty$, parameters not separately identifi-

able) being the limiting case. A pseudo-inverse would produce a small, plausible-looking SE there; withholding is the honest answer, and the ill-conditioning flag of §6.7 marks the result itself — a fit that shows the ill-conditioning flag shows no ± either. The same rule covers the float boundary at both ends: if a variance cancels to zero or below, or grows past what a double can represent, no ± is shown rather than a spuriously tight ± = 0. An unrepresentable variance withholds the ± for **every** parameter of that fit, exactly as the condition-number ceiling does — a covariance is reported whole or not at all.

6.6 How the ± is written

The displayed value's precision is set by its uncertainty (PDG convention): the SE is rounded to one significant figure — two if its leading digit is 1 — and the value is rounded to the same decimal place, e.g. $1.234\times10^6 \pm 5.6\times10^4 \rightarrow$ **$(1.23 \pm 0.06)\times10^6$**, and 40.5 ± 0.83 nm $\rightarrow$ **40.5 ± 0.8 nm**. This governs **every result display**, not just the inline ± — a value shown in a result table (the Cross-Sample ρ / R_g / R_h table, the DLS Summary R_h table, the SLS results) is rounded to the place its own SE supports, so it never implies more confidence than the ± justifies (e.g. a ±2 nm R_g is shown as `42`, not `41.7`).

Where a result has **no honest ±** (a single-correlogram size, an NNLS/CONTIN distribution peak, a single-angle M_w — §6.5 "Not reported" — and every derived number that carries no ±, including Γ, PDI, R^2 and qR_g), the value is shown at a modest **fixed number of significant figures** (default 3), set under *Settings → Result Formatting → No-uncertainty precision*. This one knob governs the ±-less results uniformly.

The sig-fig behavior, in three tiers:

1. **Exports carry full precision.** Rounding to the appropriate figures is done by the analyst at the end of a calculation chain, not by the program mid-stream.
2. **A value with an uncertainty** is displayed at the place its ± supports (the PDG rule above), so it is not shown as more certain than the ± justifies.
3. **A value with no uncertainty** is displayed at the no-uncertainty knob's precision.

A ±-less value has no uncertainty to set its precision, so its digit count is a display choice, not a claim of accuracy — the number's honesty is carried by its apparent / ±-omitted qualifier (§6.5), and the knob never touches the uncertainty-driven place of a value that *does* carry a ±.

> **Note.** Every analysis that reports an uncertainty documents its formula here with a numbered equation, so a ± you see in the program traces back to the equation behind it.

6.7 Numerical health: the ill-conditioning flag

An extrapolation design can be *numerically* degenerate while remaining physically sensible — near-duplicate concentrations, or angles clustered so tightly that $[1,\ q^2,\ c]$ is close to collinear. The platform measures this with the **condition number of the column-normalized design**, $\kappa = \mathrm{cond}(\hat{X})$, each column scaled to unit norm so that κ reflects genuine collinearity and not the arbitrary units of q^2 and c:

$$\kappa = \frac{\sigma_{\max}(\hat{X})}{\sigma_{\min}(\hat{X})}, \qquad \text{flag when } \kappa > \kappa_{\text{limit}} = 10^{12}. \tag{102}$$

A well-posed design sits at $\kappa \approx O(1\text{–}100)$, so the limit is far above anything healthy; it is a numerical criterion, not a physics one, and does not gate on the magnitude of M_{w}. When it fires the value is still shown — M_{w} and absolute A_2 are marked **unreliable**, in the SLS tab and the export Comments cell, with the advice to *add concentrations or angles, or widen their spread.* R_{g}, a slope/intercept ratio, is not gated. Reaching this state takes a genuinely collapsed ladder; it is a backstop rather than a routine warning. A fit whose M_{w} overflows to non-finite — a Guinier intercept above ≈ 710, or a Zimm intercept underflowing to zero at an extreme intensity scale — is flagged unreliable by the same rule, rather than reported as a finite-looking number.

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7. Synthetic Data Generation

The built-in generator (*Utilities* ▶ *Synthetic Generator*) exists so every analysis can be checked against a known answer. Its forward model is the same physics the analysis uses — Stokes–Einstein, q, the Siegert relation — so what it is *not* modeling matters as much as what it is. Two aspects of a real correlator's output are modeled here.

7.1 The multi-tau lag axis

A real correlator does not sample delay time on an evenly log-spaced grid. It runs a bank of linear correlators whose sample time doubles from block to block, giving a **quasi-logarithmic staircase** that reaches nine decades of lag in a few hundred channels [29, Sec. 3.2], p. 300. With L channels per octave and a finest sample time Δt_0, block j has sample time Δt_{j} and covers the lags

$$\Delta t_j = 2^j\,\Delta t_0, \qquad \tau_{j,k} = k\,\Delta t_j, \qquad k = L,\,\ldots,\,2L-1 \quad (j \geq 1), \tag{103}$$

with block 0 running $k = 1$ … $2L{-}1$. The blocks are contiguous: block $j{-}1$ ends at $(2L{-}1)\Delta t_{\mathrm{j}-1}$ and block j begins one fine step later. The $\tau = 0$ channel is omitted, since no instrument export ships one.

This matters for validation rather than for physics. Every instrument the program supports emits such an axis, so a distribution method exercised only on a uniform grid has been tested on data systematically easier than any measurement.

Some instruments report each block-boundary lag **twice**, once from each adjacent block. That is an export that has not applied the discard the multi-tau scheme prescribes [79], Eq. (11), p. 4014; the generator can reproduce it so the loader's discard has synthetic coverage.

7.2 Per-channel noise

Correlogram noise is not one number. The variance of $g_2(\tau){-}1$ varies channel by channel, and the platform's own residual weighting assumes otherwise. Taking Schätzel's per-lag variance in its **classical limit** — the mean count rate $\langle n \rangle \to \infty$, so the shot and cross terms vanish and only the intensity-noise group survives [29], Eq. (49), p. 296 — gives, with $\Gamma = \bar{\gamma}\,\Delta t_k$ the decay per sample time, k the channel index within its block and $M_{\mathrm{k}} = T/\Delta t_{\mathrm{k}}$ the number of products averaged:

$$\operatorname{var}\left(g_2(\tau_k)-1\right)=\frac{\beta^2}{M_k}\Big[\coth 2\Gamma+e^{-4\Gamma k}(2k+\coth 2\Gamma) \\ +2\beta e^{-4\Gamma k}(-2k+\coth 2\Gamma-2\coth\Gamma)+4\beta e^{-2\Gamma k}\coth\Gamma\Big]. \tag{104}$$

The resulting noise field is **not monotonic**: it falls through the correlogram's decay, then turns up sharply in the coarsest blocks, where M_{k} leaves few products to average [80, p. 29381]. What these sources corroborate is the **shape** — the non-monotonic profile and the last-stage upturn — not a magnitude: this model predicts only a *relative* span (roughly a factor of twenty over a 22-octave axis), and its absolute scale is whatever the user sets as **Noise level**, so the absolute per-channel range measured on real data [81, p. 152] cannot confirm or contradict it. The span is what carries the physics: weighting the accumulation count alone would ramp *monotonically* by a factor of ~1400, burying the longest lags under noise larger than the signal.

The limits are worth stating plainly, because the generator is a validation tool and an overstated model is worse than an honest one:

- The run duration T is common to every channel and **cancels** from the relative shape, so the generator takes none. Only the per-channel Δt_{k} survives.
- Only the *shape* is used; the absolute scale is the user's **Noise level**, which sets the first channel's SD. An absolute prediction would need a real photon count rate and the acquisition's normalization convention, neither of which a generator can supply.
- Equation (104) is a closed form derived for $\Gamma \ll 1$ [80, p. 29378], which the coarsest blocks violate by construction. The exact-in-Γ generalisation is the canonical replacement for anything that weights *real* data; an approximate shape for generated test data is a much weaker claim.
- The model is grounded on the **multi-tau** axis, where Δt_{k} is a block's real accumulation time. The generator also allows it on a uniform *log* or *linear* axis, where Δt_{k} is only the local channel spacing — so $M_{\mathrm{k}} = T/\Delta t_{\mathrm{k}}$ and "the coarsest blocks" have no physical referent there. That combination is legal and useful for exercising the code path, but the resulting noise is a shape, not a model of any correlator.

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